Методичка по ДЕМОЭКЗАМЕНУ 2022

1. Задаем имена и IP-адреса устройствам:

RTR-L:

```
router(config)#hostname RTR-L
RTR-L(config)#int g1
RTR-L(config-if)#ip add 4.4.4.100 255.255.255.0
RTR-L(config-if)#no sh
```

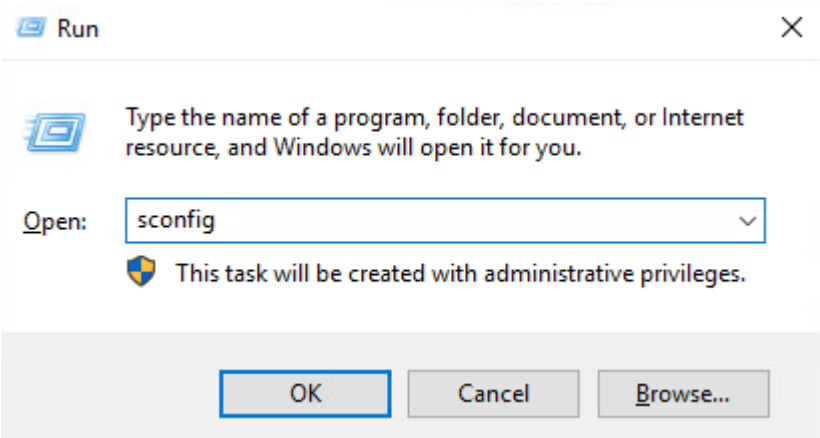
```
RTR-L(config)#int g2
RTR-L(config-if)#ip add 192.168.100.254 255.255.255.0
RTR-L(config-if)#no sh
RTR-L(config-if)#exit
```

RTR-R:

```
router(config)#hostname RTR-R
RTR-R(config)#int g1
RTR-R(config-if)#ip add 5.5.5.100 255.255.255.0
RTR-R(config-if)#no sh
RTR-R(config-if)#exit
```

```
RTR-R(config)#int g2
RTR-R(config-if)#ip add 172.16.100.254 255.255.255.0
RTR-R(config-if)#no sh
RTR-R(config-if)#exit
```

Windows Server:

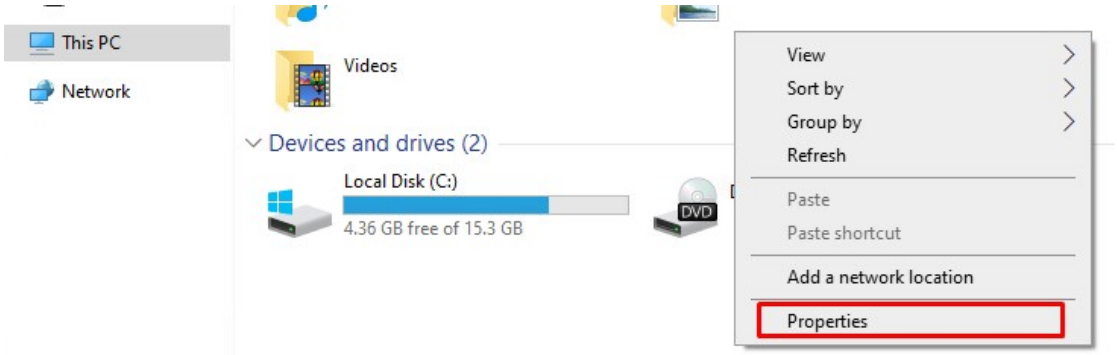


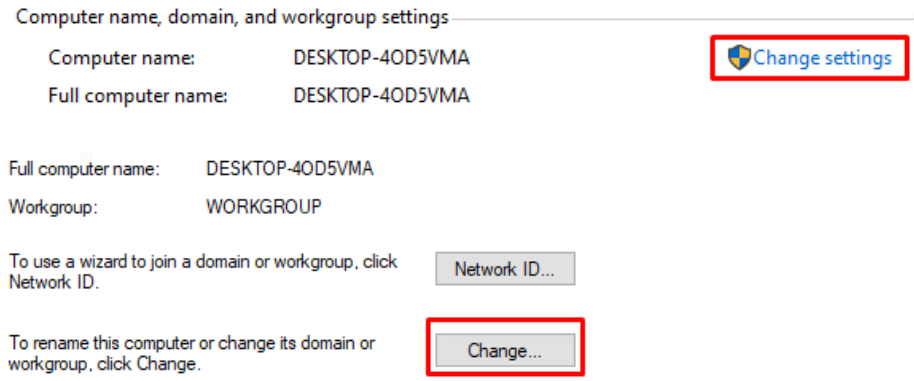

```
5) Windows Update Settings:          DownloadOnly
6) Download and Install Updates
7) Remote Desktop:                   Disabled

8) Network Settings
9) Date and Time
10) Telemetry settings               Unknown
11) Windows Activation
```

```
Available Network Adapters

Index#  IP address      Description

 1      192.168.100.200 Intel(R) 82574L Gigabit Network Connection
```

```
C:\Windows\system32\cmd.exe
Microsoft (R) Windows Script Host Version 5.812
Copyright (C) Microsoft Corporation. All rights reserved.

Inspecting system...


================================================================================
                          Server Configuration
================================================================================

1) Domain/Workgroup:                    Workgroup:  WORKGROUP
2) Computer Name:                       WIN-OAL0M2EAHN4
3) Add Local Administrator
4) Configure Remote Management          Enabled
```

Windows:

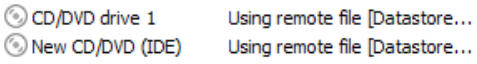




Как поставить IP-адрес на винде – сами знаете.

На ISP:

- Добавляем еще 1 дисковод и в каждый дисковод вставляем диски из хранилища (BD1 и BD2):



- Добавляем диски в самой системе командой:

```
root@debian:~# apt-cdrom add
```

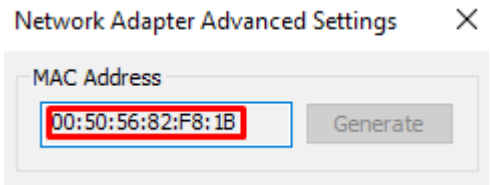
- Запускаем установку пакетов:

```
root@debian:~# apt install vim network-manager bind9 chrony -y
```

```
root@debian:~# nmtui
```

Сравниваем mac-адреса адаптеров с mac-адресами интерфейсов:

```
Device ens192 (00:50:56:82:F8:1B)
```



Назначаем IP-адреса:

```
IPv4 CONFIGURATION <Manual>
         Addresses 4.4.4.1/24_____________ <Remove>
                   <Add...>
           Gateway ________________________
       DNS servers <Add...>
   Search domains  <Add...>
```

```
IPv4 CONFIGURATION <Manual>
         Addresses 5.5.5.1/24_____________ <Remove>
                   <Add...>
           Gateway ________________________
       DNS servers <Add...>
   Search domains  <Add...>
```

!!! Внимание!!! На ISP, на адресе 3.3.3.1 или на любом другом назначаем DNS!

```
IPv4 CONFIGURATION <Manual>
         Addresses 3.3.3.1/24_____________ <Remove>
                   <Add...>
           Gateway ________________________
       DNS servers 127.0.0.1_______________ <Remove>
                   <Add...>
```

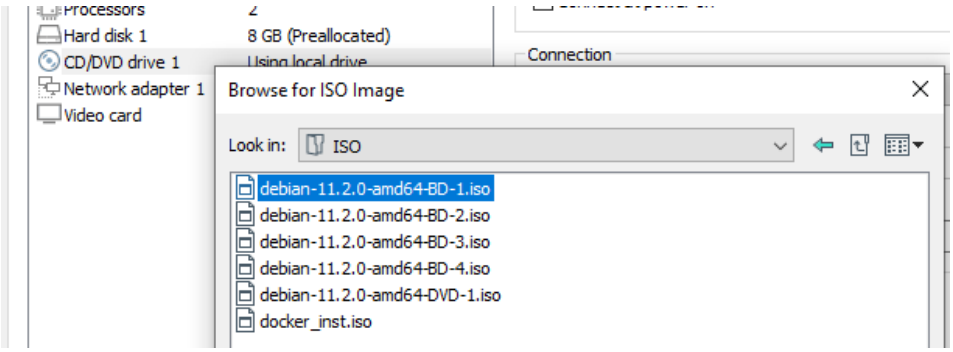
В этой же программе задаем имя.

```
NetworkManager TUI

Please select an option

Edit a connection
Activate a connection
Set system hostname

Quit

                    <OK>
```

```
Set Hostname

Hostname ISP_____________________________________

                                  <Cancel> <OK>
```

На машинах WEB:

- Добавляем в дисковод первый диск из хранилища:



- Добавляем диски в самой системе командой:

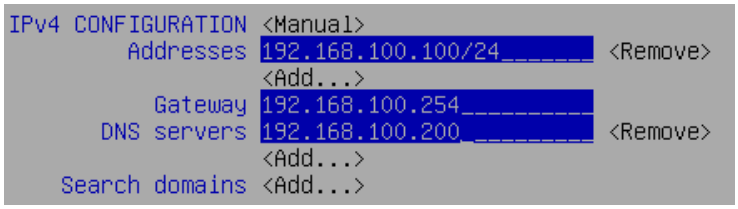
```
root@debian:~# apt-cdrom add
```

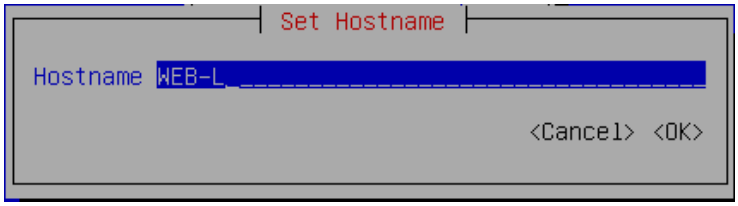
- Запускаем установку пакетов:

```
root@debian:~# apt install vim network-manager chrony openssh-server ssh -y
```

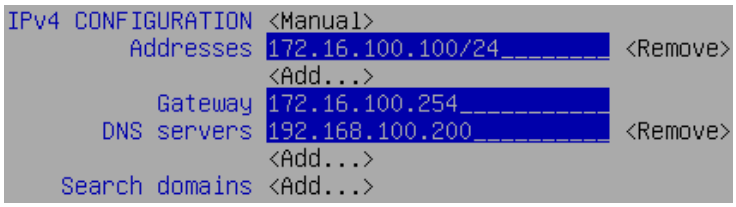
- Назначаем IP-адреса и имена:

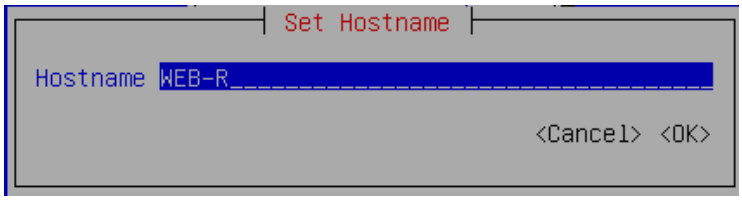
Web-L:





Web-R:





2. Настраиваем шлюз последней надежды:

```
RTR-L(config)#ip route 0.0.0.0 0.0.0.0 4.4.4.1
```

```
RTR-R(config)#ip route 0.0.0.0 0.0.0.0 5.5.5.1
```

3. Включаем хождение трафика на ISP:

```
root@debian:~# vim /etc/sysctl.conf
```

Убираем знак решетки перед фразой

```
net.ipv4.ip_forward=1
```

Применяем параметры

```
root@debian:~# sysctl -p
net.ipv4.ip_forward = 1
```

4. Настройка NAT

RTR-L:

Создаем ACL-список и указываем внутренний адрес для его трансляции во внешний:

```
RTR-L(config)#access-list 1 permit 192.168.100.0 0.0.0.255
RTR-L(config)#access-list 1 deny any
```

Указываем, какой интерфейс внутренний, а какой внешний (входящий и исходящий):

```
RTR-L(config)#int g1
RTR-L(config-if)#ip nat outside
RTR-L(config-if)#exit
RTR-L(config)#int g2
RTR-L(config-if)#ip nat inside
RTR-L(config-if)#exit
```

Настраиваем NAT с перегрузкой (PAT), для этого в конце пишем overload:

```
RTR-L(config)#ip nat inside source list 1 interface g1 overload
```

RTR-R:

```
RTR-R(config)#access-list 1 permit 172.16.100.0 0.0.0.255
RTR-R(config)#access-list 1 deny any
RTR-R(config)#int g1
RTR-R(config-if)#ip nat outside
RTR-R(config-if)#exit
RTR-R(config)#int g2
RTR-R(config-if)#ip nat inside
RTR-R(config-if)#exit
RTR-R(config)#ip nat inside source list 1 interface g1 overload
```

5. Настройка туннеля GRE+IPSEC

RTR-L:

Создаем политику для IPSec (первая фаза аутентификации):

```
RTR-L(config)#crypto isakmp policy 1
RTR-L(config-isakmp)#encryption aes
RTR-L(config-isakmp)#authentication pre-share
RTR-L(config-isakmp)#hash sha256
RTR-L(config-isakmp)#group 14
RTR-L(config-isakmp)#exit
```

Создаем ключ для аутентификации IPSec и настраиваем хождение через NAT:

```
RTR-L(config)#crypto isakmp key SuperSecretKey address 5.5.5.100
RTR-L(config)#crypto isakmp nat keepalive 5
```

Создаем transform-set для IPSec с режимом туннеля (вторая фаза аутентификации):

```
RTR-L(config)#crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
RTR-L(cfg-crypto-trans)#mode tunnel
RTR-L(cfg-crypto-trans)#exit
```

Создаем профиль для защиты IPSec туннеля, используя раннее созданный transform-set:

```
RTR-L(config)#crypto ipsec profile ipsec
RTR-L(ipsec-profile)#set transform-set TSET
RTR-L(ipsec-profile)#exit
```

Настраиваем сам ipsec-туннель и в качестве защиты устанавливаем раннее созданный профиль:

```
RTR-L(config)#int tun1
RTR-L(config-if)#tunnel source 4.4.4.100
RTR-L(config-if)#tunnel destination 5.5.5.100
RTR-L(config-if)#ip add 10.8.8.1 255.255.255.252
RTR-L(config-if)#tunnel mode ipsec ipv4
RTR-L(config-if)#tunnel protection ipsec profile ipsec
```

RTR-R:

Настройка аналогична, за исключением адреса назначения

```
RTR-R(config)#crypto isakmp policy 1
RTR-R(config-isakmp)#encryption aes
RTR-R(config-isakmp)#authentication pre-share
RTR-R(config-isakmp)#hash sha256
RTR-R(config-isakmp)#group 14
RTR-R(config-isakmp)#exit
```

```
RTR-R(config)#crypto isakmp key SuperSecretKey address 4.4.4.100
RTR-R(config)#crypto isakmp nat keepalive 5
```

```
RTR-R(config)#crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
RTR-R(cfg-crypto-trans)#mode tunnel
RTR-R(cfg-crypto-trans)#exit_
```

```
RTR-R(config)#crypto ipsec profile ipsec
RTR-R(ipsec-profile)#set transform-set TSET
RTR-R(ipsec-profile)#exit
```

```
RTR-R(config)#int tun1
RTR-R(config-if)#tunnel source 5.5.5.100
RTR-R(config-if)#tunnel destination 4.4.4.100
RTR-R(config-if)#ip address 10.8.8.2 255.255.255.252
RTR-R(config-if)#tunnel mode ipsec ipv4
RTR-R(config-if)#tunnel protection ipsec profile ipsec
```

Если появится такое сообщение, значит туннель работает:

```
*Jun  5 14:18:02.561: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, c
hanged state to up
```

6. Настройка маршрутизации между внутренними сетями по туннелю (можно использовать любой протокол)

RTR-L:

```
RTR-L(config)#router eigrp 65000
RTR-L(config-router)#network 192.168.100.0 0.0.0.255
RTR-L(config-router)#network 10.8.8.0 0.0.0.3
```

RTR-R:

```
RTR-R(config)#router eigrp 65000
RTR-R(config-router)#network 172.16.100.0 0.0.0.255
RTR-R(config-router)#network 10.8.8.0 0.0.0.3
RTR-R(config-router)#
*Jun  5 14:24:00.443: %DUAL-5-NBRCHANGE: EIGRP-IPv4 65000: Neighbor 10.8.8.1 (Tu
nnel1) is up: new adjacency
```

Как видим, сосед найден, а значит все настроено правильно.

7. Настройка ACL-списков

RTR-L:

```
RTR-L(config)#ip access-list extended Left
RTR-L(config-ext-nacl)#permit tcp any any established
RTR-L(config-ext-nacl)#permit udp host 4.4.4.1 eq 53 any
RTR-L(config-ext-nacl)#permit tcp any host 4.4.4.100 eq 80
RTR-L(config-ext-nacl)#permit tcp any host 4.4.4.100 eq 443
RTR-L(config-ext-nacl)#permit tcp any host 4.4.4.100 eq 2222
RTR-L(config-ext-nacl)#permit udp host 4.4.4.1 eq 123 any
RTR-L(config-ext-nacl)#permit udp host 5.5.5.100 host 4.4.4.100 eq 500
RTR-L(config-ext-nacl)#permit esp any any
RTR-L(config-ext-nacl)#permit icmp any any
RTR-L(config-ext-nacl)#exit
```

```
RTR-L(config)#int g1
RTR-L(config-if)#ip access-group Left in
```

RTR-R:

```
RTR-R(config)#ip access-list extended Right
RTR-R(config-ext-nacl)#permit tcp any any established
RTR-R(config-ext-nacl)#permit tcp any host 5.5.5.100 eq 80
RTR-R(config-ext-nacl)#permit tcp any host 5.5.5.100 eq 443
RTR-R(config-ext-nacl)#permit tcp any host 5.5.5.100 eq 2244
RTR-R(config-ext-nacl)#permit udp host 4.4.4.100 host 5.5.5.100 eq 500
RTR-R(config-ext-nacl)#permit esp any any
RTR-R(config-ext-nacl)#permit icmp any any
```

```
RTR-R(config)#int g1
RTR-R(config-if)#ip access-group Right in
```

8. Пробрасываем порты для подключения к серверу WEB извне:

RTR-L:

```
RTR-L(config)#ip nat inside source static tcp 192.168.100.100 22 4.4.4.100 2222
```

RTR-R:

```
RTR-R(config)#ip nat inside source static tcp 172.16.100.100 22 5.5.5.100 2244
```

9. Настраиваем ssh на машинах WEB:

Заходим в файл, чтобы разрешить подключение от пользователя root:

```
root@debian:~# vim /etc/ssh/sshd_config
```

```
#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Перезагружаем службы:

```
root@debian:~# systemctl restart ssh sshd
```

Пробуем подключиться по SSH:

```
root@ISP:~# ssh root@4.4.4.100 -p 2222
The authenticity of host '[4.4.4.100]:2222 ([4.4.4.100]:2222)' can't be established.
ECDSA key fingerprint is SHA256:MraoqsLsYTs3ShM3dOBgP/woD64NSYBcduOzNENkTb4.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[4.4.4.100]:2222' (ECDSA) to the list of known hosts.
root@4.4.4.100's password:
Linux WEB-L 5.10.0-10-amd64 #1 SMP Debian 5.10.84-1 (2021-12-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Apr  1 05:20:09 2022
root@WEB-L:~#
```

Подключение проходит.

10. Настройка DNS-сервера на машине ISP

Выключаем защиту зон:

```
root@debian:~# vim /etc/bind/named.conf.options
```

```
  dnssec-validation no;
```

Заходим в файл настройки основных зон DNS:

```
root@ISP:~# vim /etc/bind/named.conf.default-zones
```

Добавляем зону по заданию:

```
zone "demo.wsr" {
        type master;
        file "/opt/dns/demo.wsr";
};
```

Type master – говорит, что зона является главной.

File “” – путь к настройке зоны.

Создаем путь, который указали в настройках:

```
root@ISP:~# mkdir /opt/dns
```

Разрешаем запись в каталог:

```
root@ISP:~# vim /etc/apparmor.d/usr.sbin.named
```

```
/opt/dns/** rw,
```

```
root@ISP:~# systemctl restart apparmor.service
```

Копируем в созданный ранее каталог зону из папки bind:

```
root@ISP:~# cp /etc/bind/db.local /opt/dns/demo.wsr
```

Обратите внимание, при копировании сразу меняется имя файла!

Назначаем права доступа к папке:

```
root@ISP:~# chown -R bind:bind /opt/dns/
```

Заполняем зону в соответствии с таблицей 2:

```
$TTL    604800
demo.wsr.       IN      SOA     demo.wsr. root.demo.wsr. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
demo.wsr.       IN      NS      isp.demo.wsr.
isp             IN      A       3.3.3.1
www             IN      A       4.4.4.100
www             IN      A       5.5.5.100
internet        IN      CNAME   isp.demo.wsr.
```

NS - nameserver, то есть тут указываем наши DNS-сервера

A – address, тут мы сопоставляем имя с IP-адресом

CNAME – canonical name, тут мы делаем дополнительное имя для уже созданной записи с адресом.

Перезагружаем службу:

```
root@ISP:~# systemctl restart bind9
```

Проверяем работу dns:

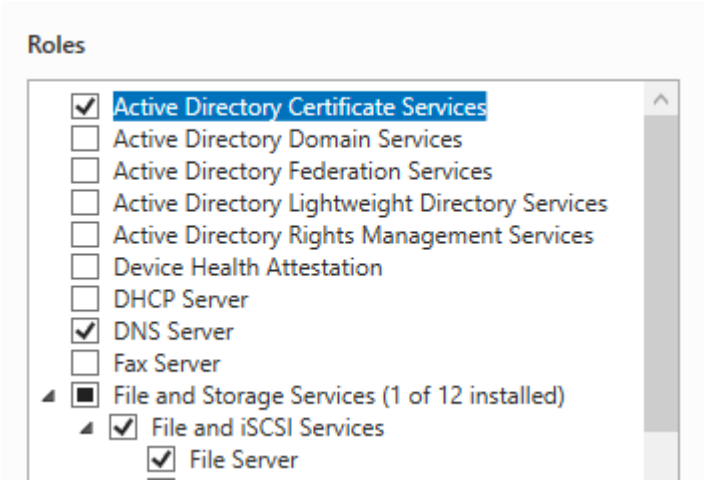
```
root@debian:~# nslookup
> www.demo.wsr
Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   www.demo.wsr
Address: 5.5.5.100
Name:   www.demo.wsr
Address: 4.4.4.100
> internet.demo.wsr
Server:         127.0.0.1
Address:        127.0.0.1#53

internet.demo.wsr       canonical name = isp.demo.wsr.
Name:   isp.demo.wsr
Address: 3.3.3.1
```

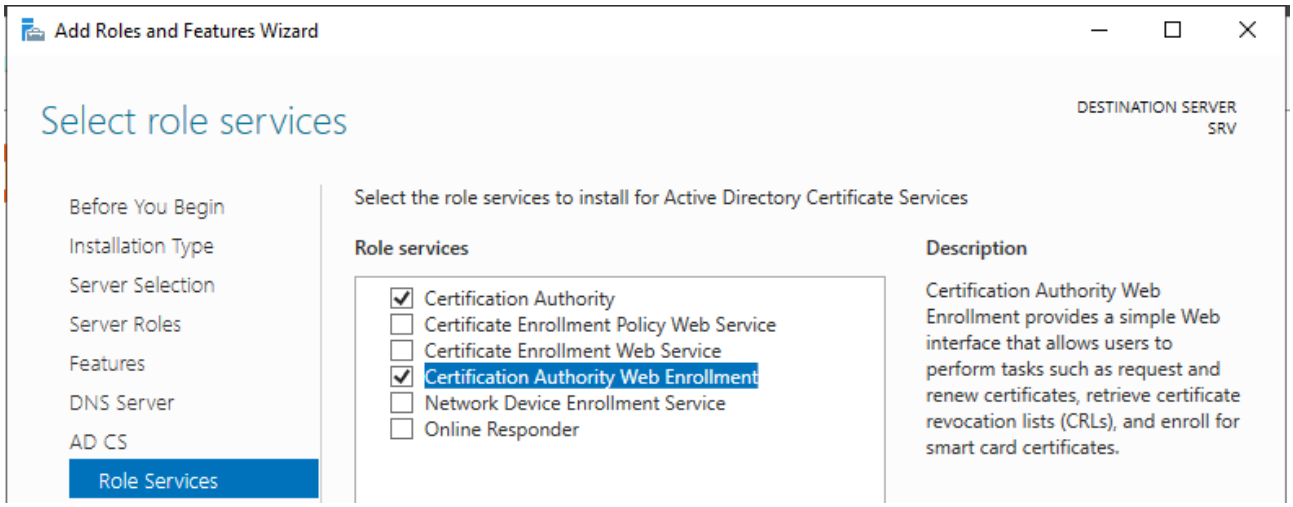
Все работает!

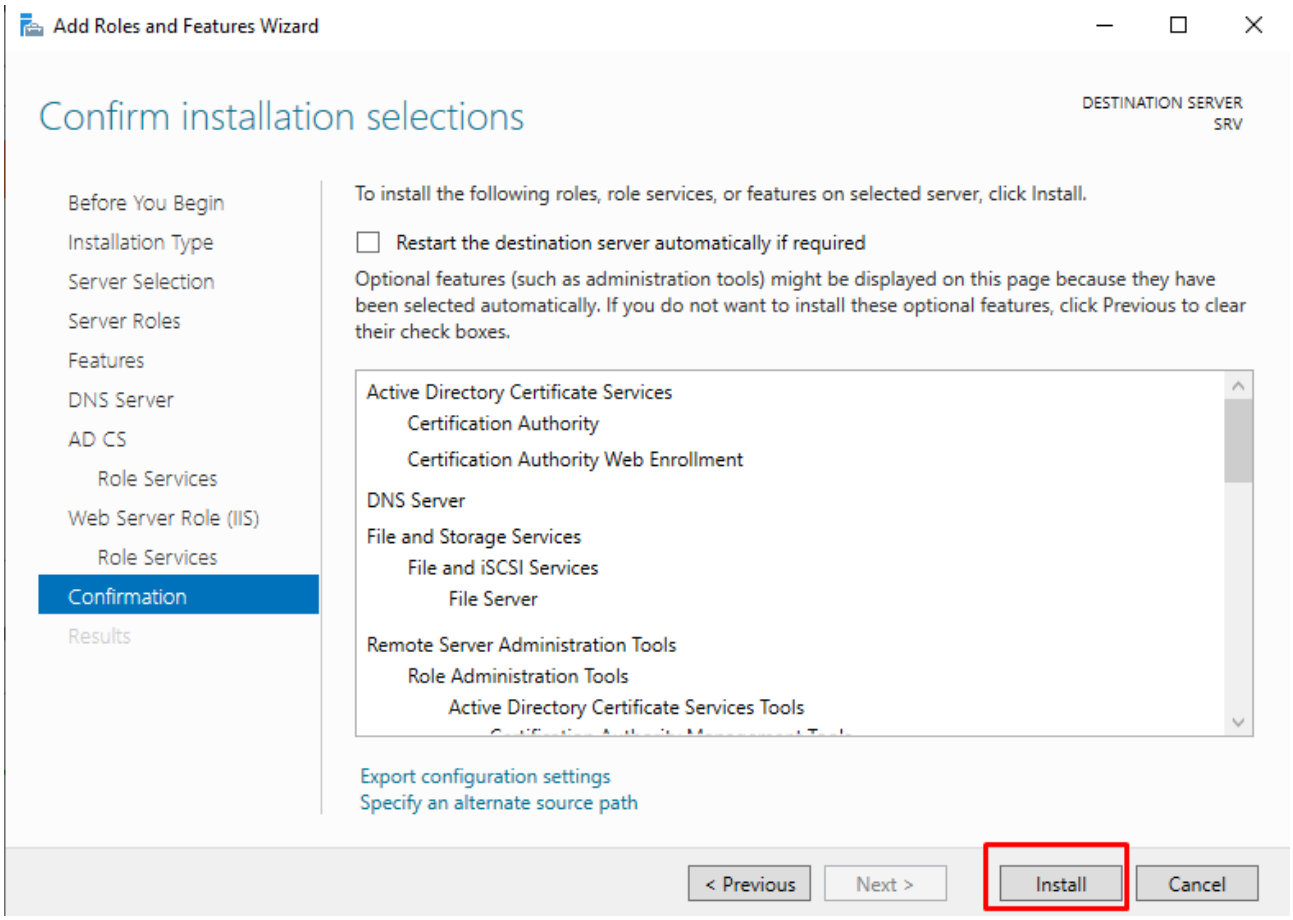
11. Маршрутизатор региона должен транслировать соответствующие порты DNS-службы в порты сервера SRV:

```
RTR-L(config)#ip nat inside source static tcp 192.168.100.200 53 4.4.4.100 53
RTR-L(config)#ip nat inside source static udp 192.168.100.200 53 4.4.4.100 53
```

12. Устанавливаем службы на машину SRV.

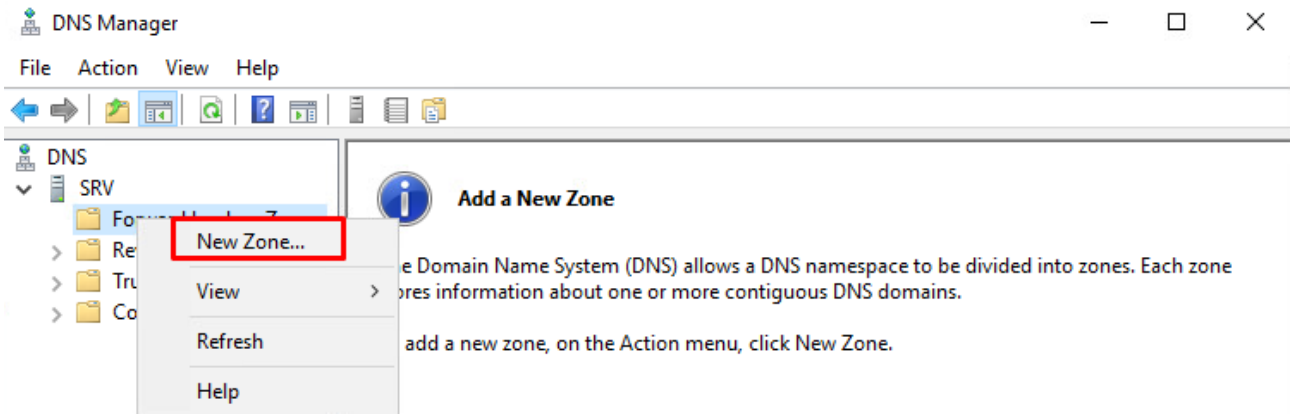

Roles
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
Fax Server
File and Storage Services (1 of 12 installed)
File and iSCSI Services
File Server

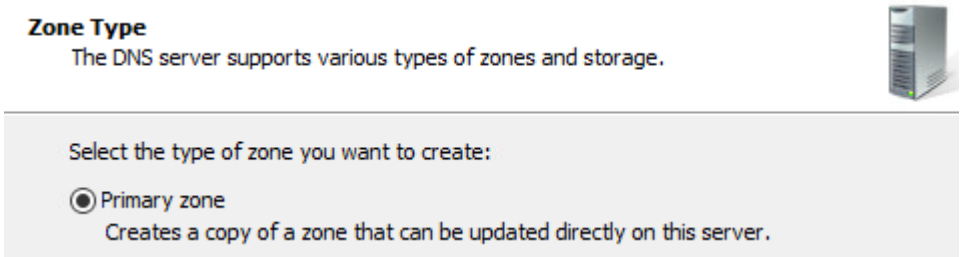

Add Roles and Features Wizard
Select role services
DESTINATION SERVER
SRV
Before You Begin
Installation Type
Server Selection
Server Roles
Features
DNS Server
AD CS
Role Services
Select the role services to install for Active Directory Certificate Services
Role services
Certification Authority
Certificate Enrollment Policy Web Service
Certificate Enrollment Web Service
Certification Authority Web Enrollment
Network Device Enrollment Service
Online Responder
Description
Certification Authority Web Enrollment provides a simple Web interface that allows users to perform tasks such as request and renew certificates, retrieve certificate revocation lists (CRLs), and enroll for smart card certificates.

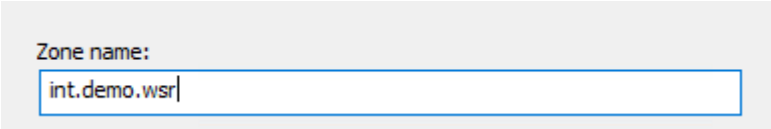

Add Roles and Features Wizard
Confirm installation selections
DESTINATION SERVER
SRV
Before You Begin
Installation Type
Server Selection
Server Roles
Features
DNS Server
AD CS
Role Services
Web Server Role (IIS)
Role Services
Confirmation
Results
To install the following roles, role services, or features on selected server, click Install.
Restart the destination server automatically if required
Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.
Active Directory Certificate Services
Certification Authority
Certification Authority Web Enrollment
DNS Server
File and Storage Services
File and iSCSI Services
File Server
Remote Server Administration Tools
Role Administration Tools
Active Directory Certificate Services Tools
Export configuration settings
Specify an alternate source path
< Previous
Next >
Install
Cancel

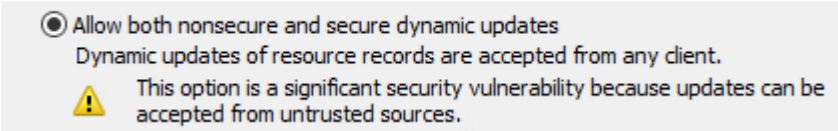
13.Настраиваем службу DNS на SRV.

Создаем прямую зону:





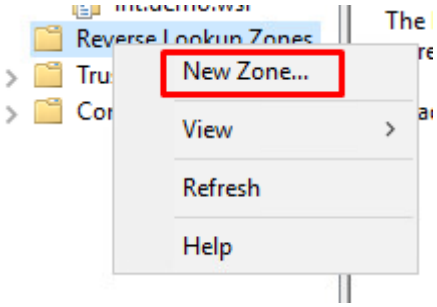


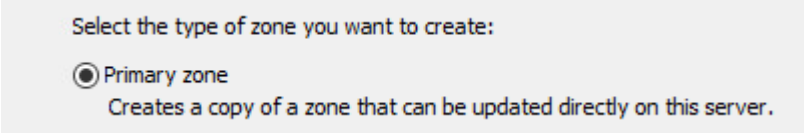


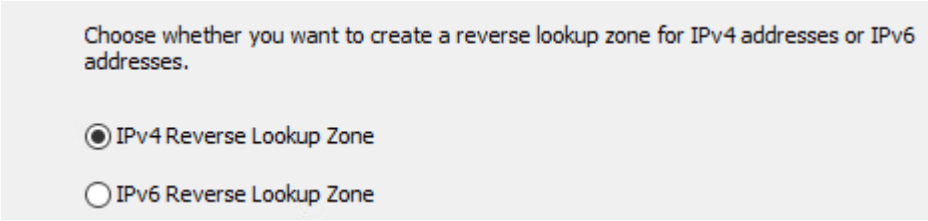
Создаем обратные зоны:

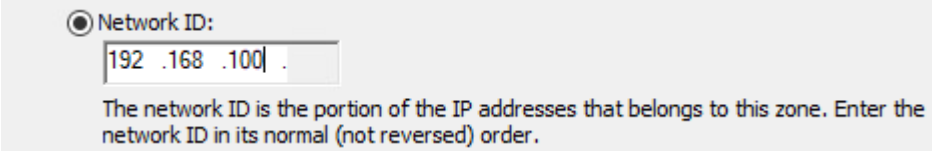
Их две: 192.168.100 и 172.16.100

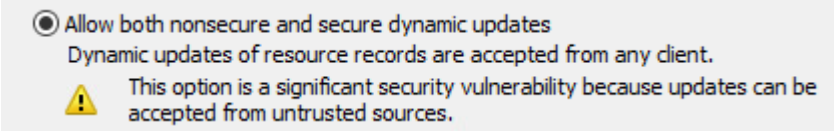
Покажу на примере одной:





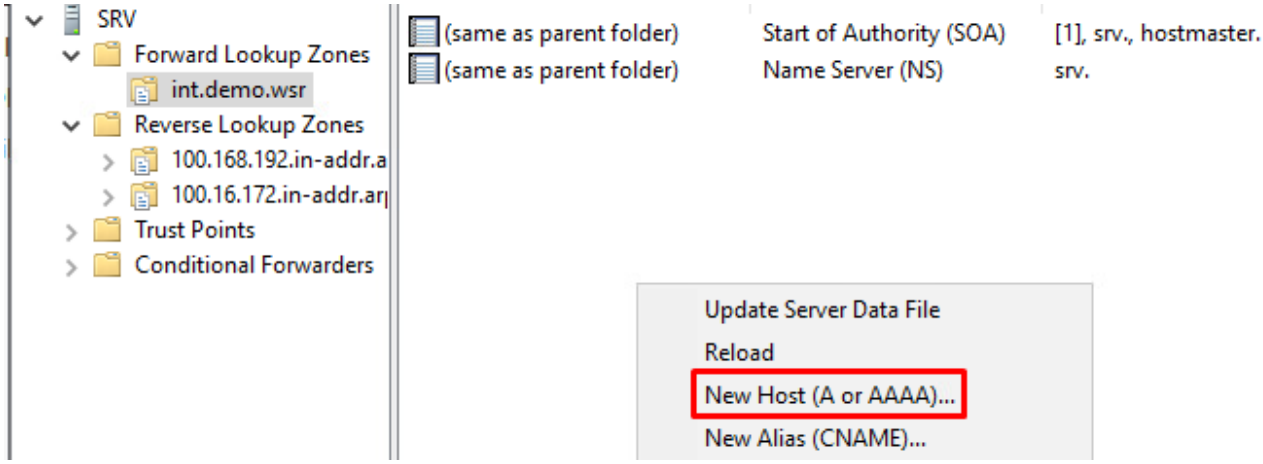


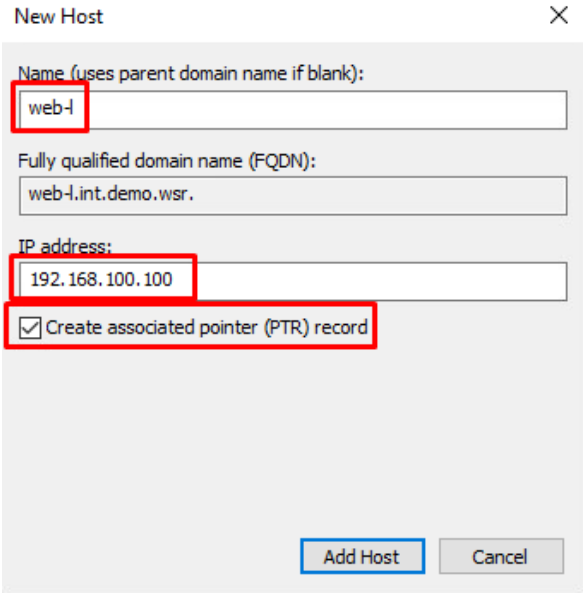




Заполняем прямую зону в соответствии с таблицей 2:

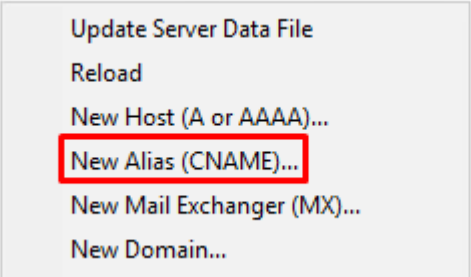
Записи А добавляются так:

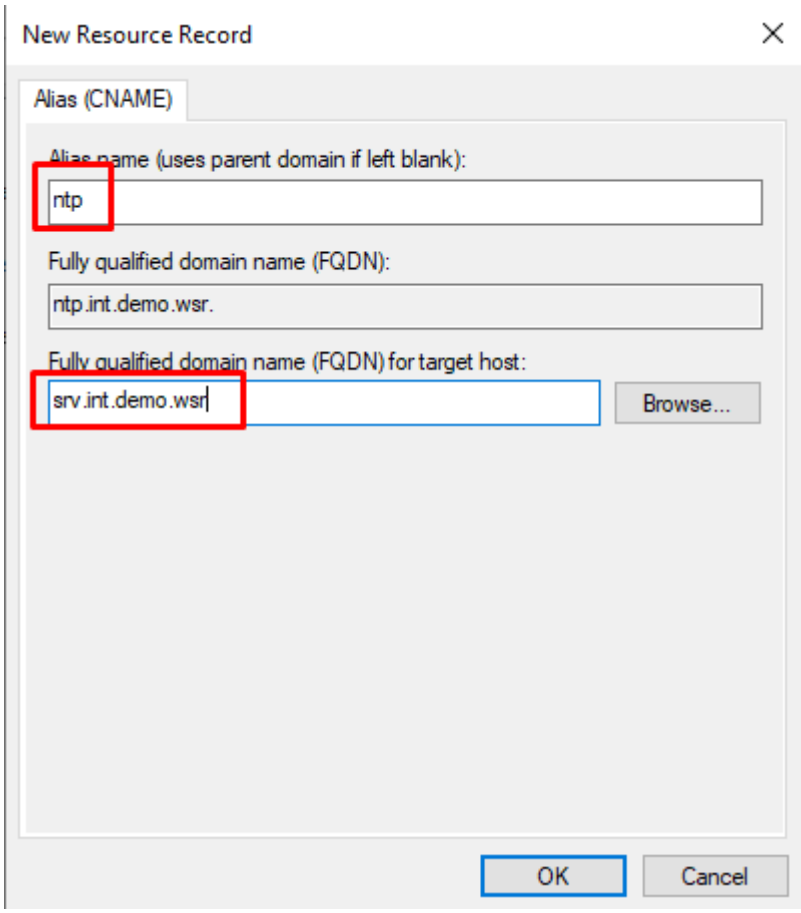




Обратите внимание, что галочка означает создание обратной записи в reverse зоне (это те две, которые мы создавали (192.168.100 и 172.16.100)).

Записи типа CNAME создаются так:

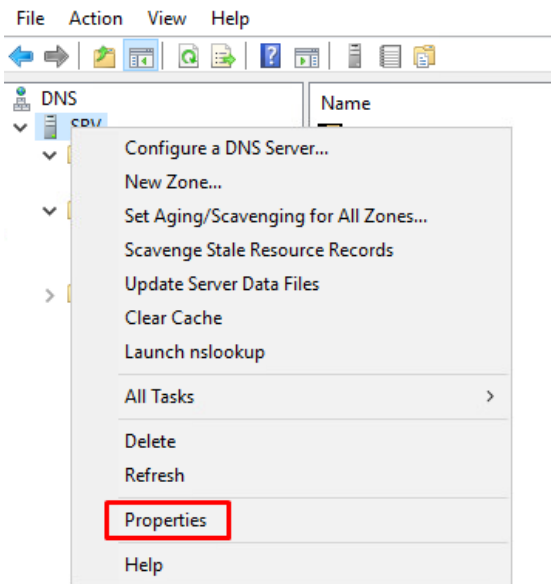


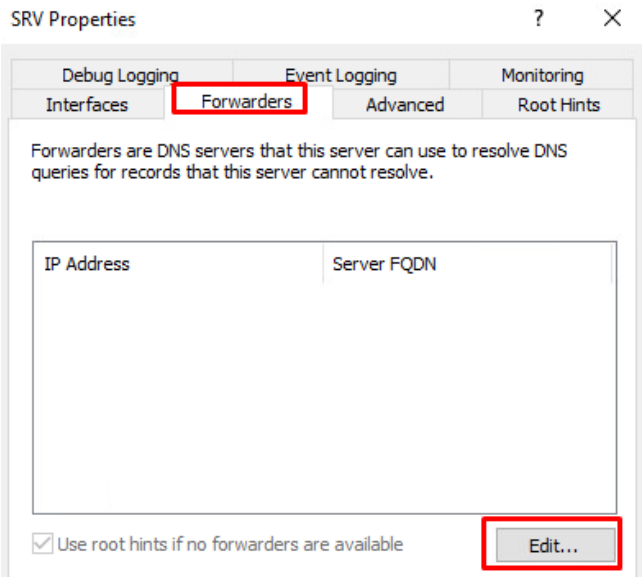


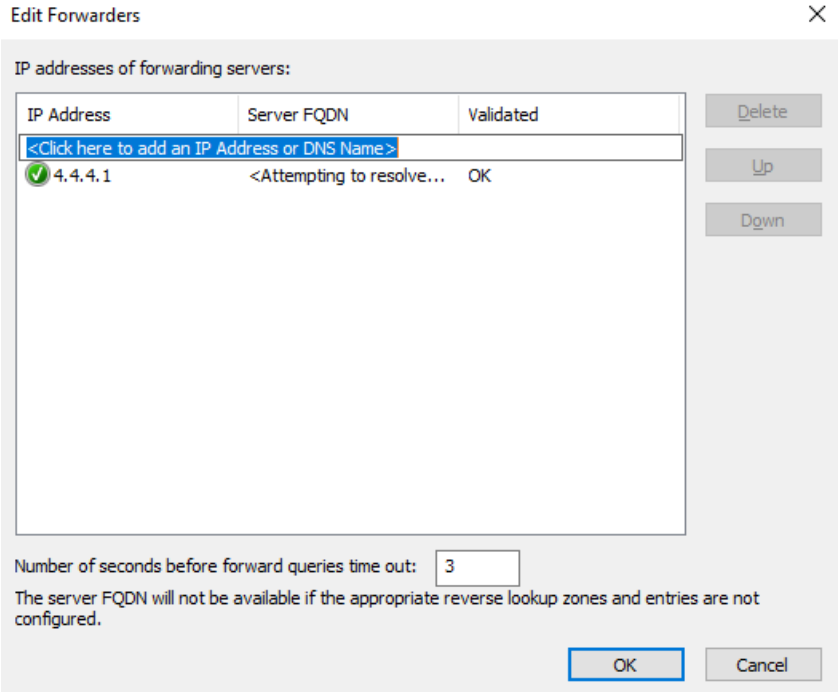
Надо заполнить все записи из таблицы 2!

web-l	Host (A)	192.168.100.100
web-r	Host (A)	172.16.100.100
rtr-l	Host (A)	192.168.100.254
rtr-r	Host (A)	172.16.100.254
srv	Host (A)	192.168.100.200
ntp	Alias (CNAME)	srv.int.demo.wsr
dns	Alias (CNAME)	srv.int.demo.wsr

Добавляем forwarder:


File
Action
View
Help
DNS
Name
Configure a DNS Server...
New Zone...
Set Aging/Scavenging for All Zones...
Scavenge Stale Resource Records
Update Server Data Files
Clear Cache
Launch nslookup
All Tasks
Delete
Refresh
Properties
Help


SRV Properties
Debug Logging
Event Logging
Monitoring
Interfaces
Forwarders
Advanced
Root Hints
Forwarders are DNS servers that this server can use to resolve DNS queries for records that this server cannot resolve.
IP Address
Server FQDN
Use root hints if no forwarders are available
Edit...


Edit Forwarders
IP addresses of forwarding servers:
IP Address
Server FQDN
Validated
<Click here to add an IP Address or DNS Name>
4.4.4.1
<Attempting to resolve...
OK
Delete
Up
Down
Number of seconds before forward queries time out:
3
The server FQDN will not be available if the appropriate reverse lookup zones and entries are not configured.
OK
Cancel

Forwarder нужен для того, чтобы отправлять ему все не известные данному серверу записи.

Проверяем работу сервера:

```
root@debian:~# nslookup
> ntp.int.demo.wsr
Server:         192.168.100.200
Address:        192.168.100.200#53

ntp.int.demo.wsr        canonical name = srv.int.demo.wsr.
Name:   srv.int.demo.wsr
Address: 192.168.100.200
> internet.demo.wsr
Server:         192.168.100.200
Address:        192.168.100.200#53

Non-authoritative answer:
internet.demo.wsr       canonical name = isp.demo.wsr.
Name:   isp.demo.wsr
Address: 3.3.3.1
```

Работает!

14. Настройка NTP на машине ISP:

```
root@ISP:~# vim /etc/chrony/chrony.conf
```

```
pool 2.debian.pool.ntp.org iburst

local stratum 4
allow 3.3.3.0/24
allow 4.4.4.0/24
```

Pool – уже настроен по умолчанию.

Local stratum – максимальное количество переходов между машинами, а также говорит системе о том, что она считает свое время правильным.

Allow – с какими сетями работаем.

Устанавливаем нашу временную зону:

```
root@ISP:~# timedatectl set-timezone Europe/Moscow
```

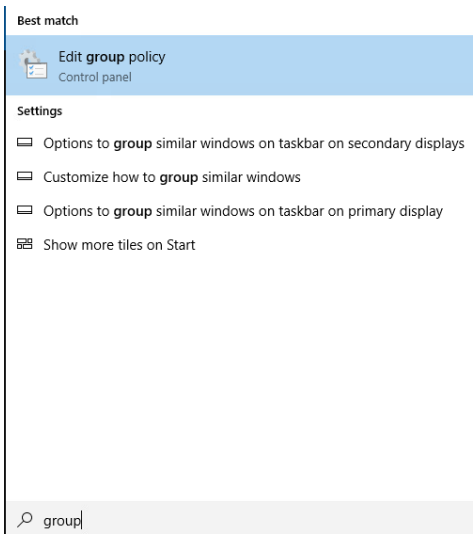
Включаем и перезагружаем службу синхронизации времени:

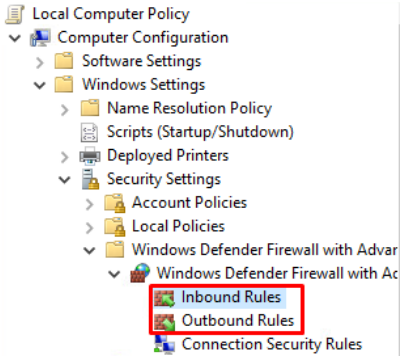
```
root@ISP:~# systemctl enable chrony
Synchronizing state of chrony.service with SysV service script with /lib/systemd/systemd-sysv-instal
l.
Executing: /lib/systemd/systemd-sysv-install enable chrony
root@ISP:~# systemctl restart chrony
```

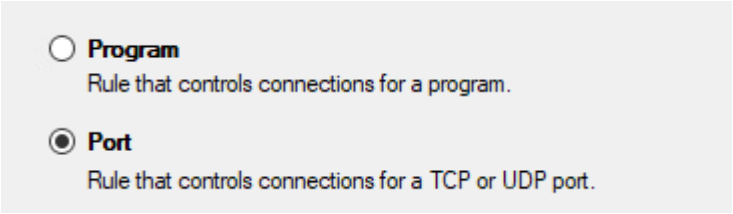
Проверяем:

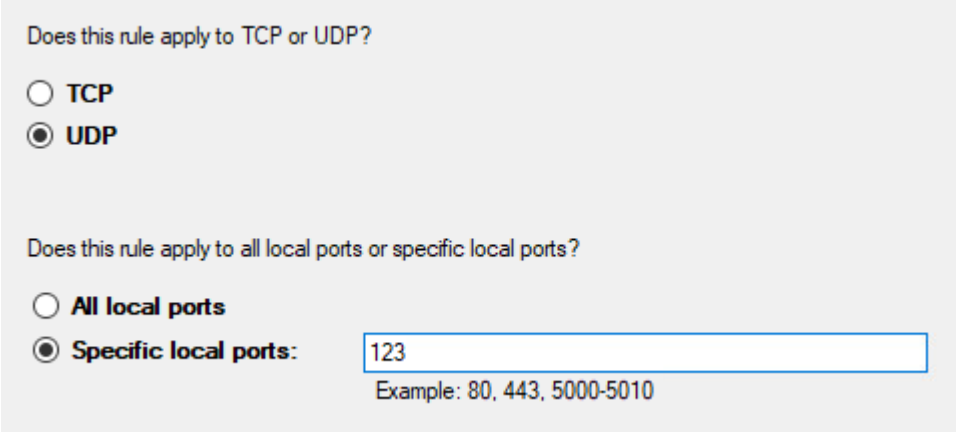
```
root@ISP:~# date
Mon 04 Apr 2022 11:40:43 AM MSK
```

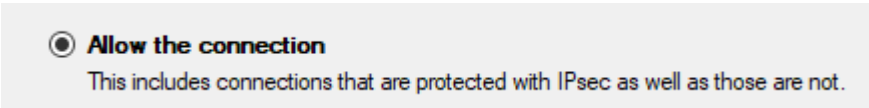
15.Настройка NTP на машине SRV:

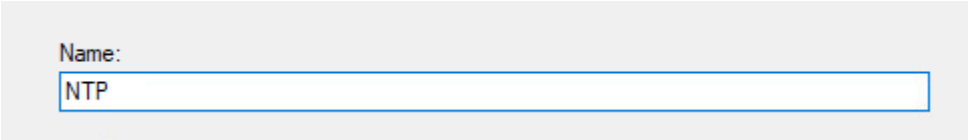


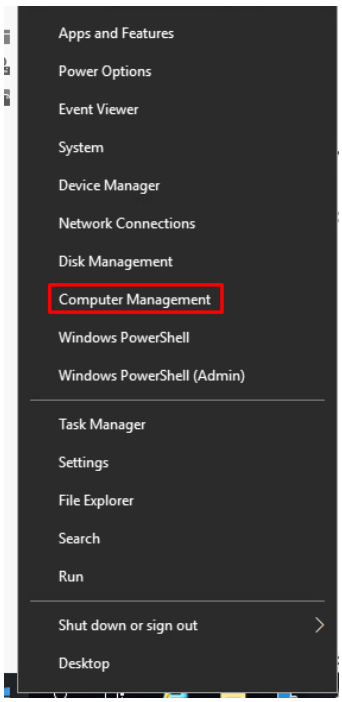


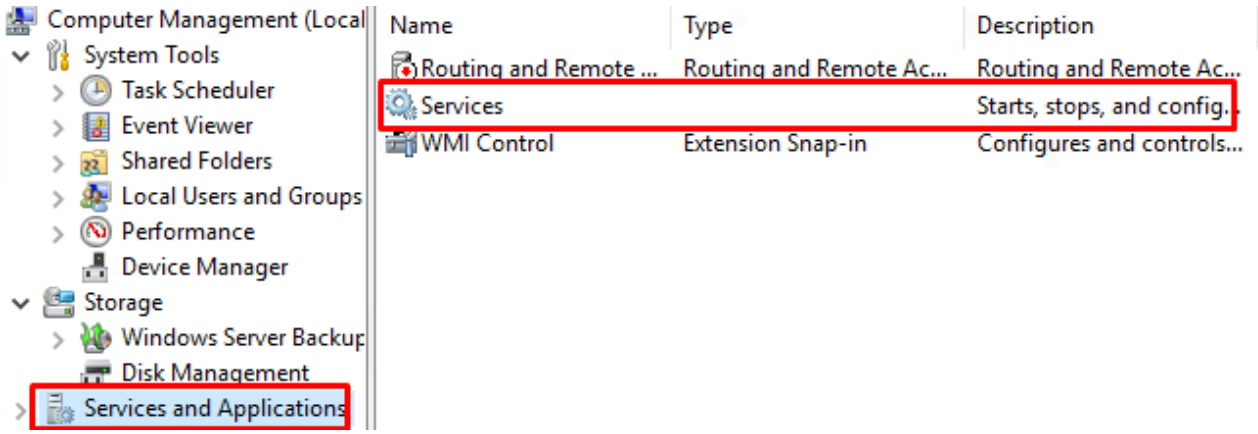


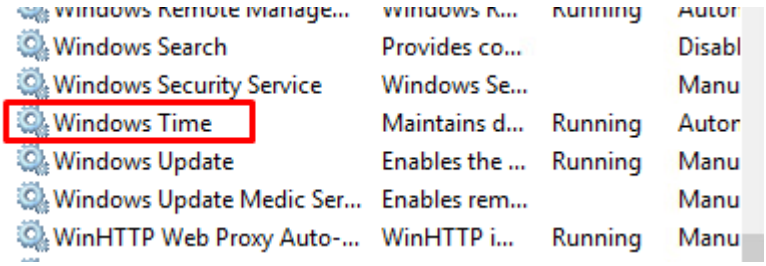






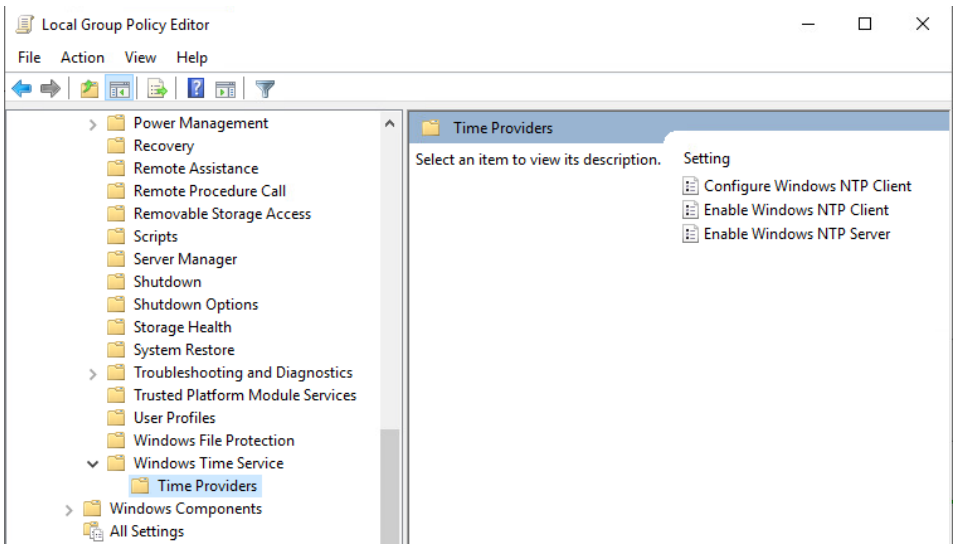




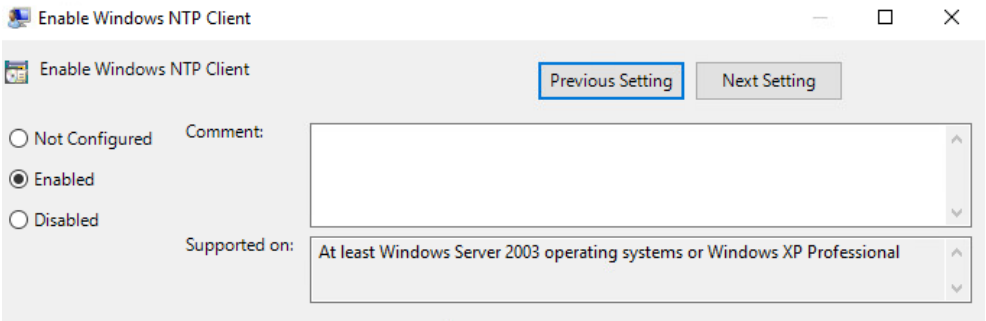


Настраиваем NTP:

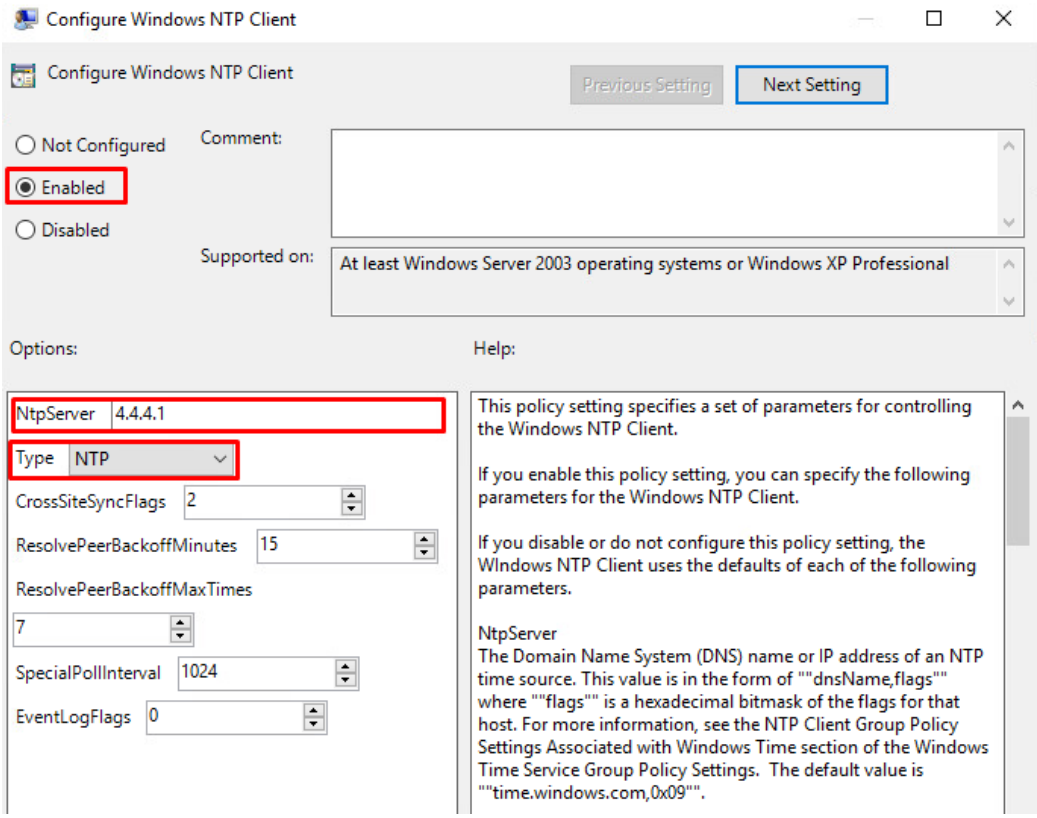
Переходим в редактор групповой политики по пути **Computer Configuration -> Administrative Templates -> System -> Windows Time Service -> Time Providers**



Переводим групповые политики **Enable Windows NTP Client** и **Enable Windows NTP Server** в режим **Enabled**.



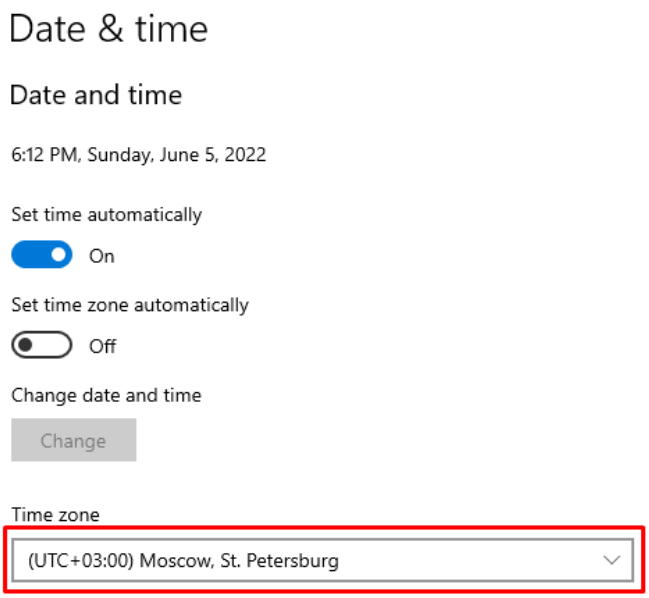
Настраиваем Windows NTP Client:



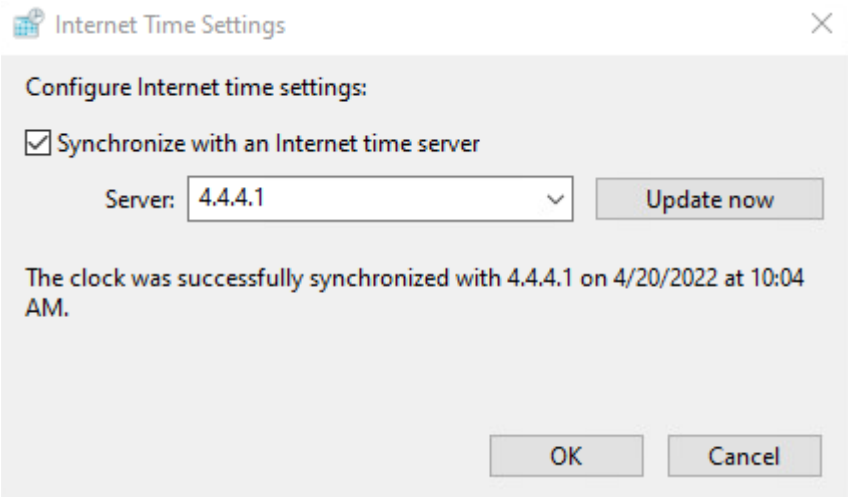
Обновляем групповую политику:

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.


C:\Users\Administrator>
```

Настраиваем временную зону:



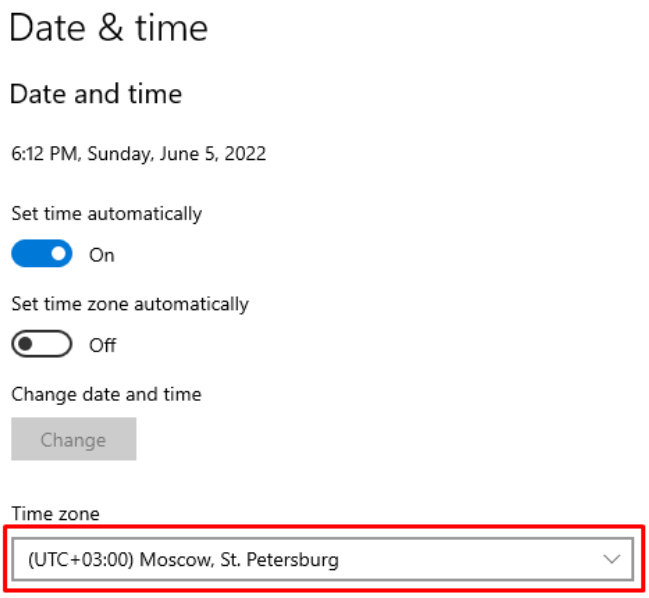
Синхронизируем время:



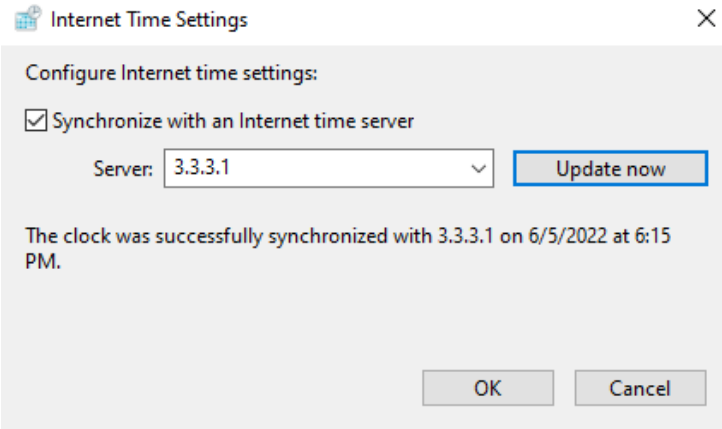
16. Настраиваем NTP на остальных машинах

CLI:

Настраиваем временную зону:



Синхронизируем время:



Машины WEB:

```
root@debian:~# vim /etc/chrony/chrony.conf
```

```
pool ntp.int.demo.wsr iburst
allow 192.168.100.0/24_
```

```
timedatectl set-timezone Europe/Moscow
```

```
root@debian:~# systemctl enable chrony
Synchronizing state of chrony.service with SysV service script with /lib/systemd/systemd-sysv-instal
l.
Executing: /lib/systemd/systemd-sysv-install enable chrony
root@debian:~# systemctl restart chrony
```

```
root@debian:~# timedatectl show
Timezone=Europe/Moscow
LocalRTC=no
CanNTP=yes
NTP=yes
NTPSynchronized=yes
TimeUSec=Sun 2022-06-05 18:21:14 MSK
RTCTimeUSec=Sun 2022-06-05 18:21:14 MSK
```

Cisco:

```
RTR-L(config)#clock timezone UTC +3
RTR-L(config)#ip domain name int.demo.wsr
RTR-L(config)#ip name-server 192.168.100.200
RTR-L(config)#ntp server ntp.int.demo.wsr
```

```
RTR-R(config)#clock timezone UTC +3
RTR-R(config)#ip domain name int.demo.wsr
RTR-R(config)#ip name-server 192.168.100.200
RTR-R(config)#ntp server ntp.int.demo.wsr
```

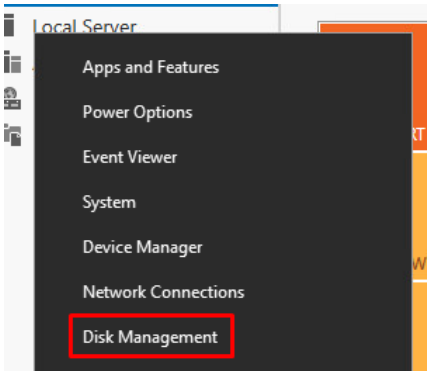
На данных машинах время синхронизируется минут через 10:

```
RTR-L#show ntp status
Clock is synchronized, stratum 6, reference is 192.168.100.200
nominal freq is 250.0000 Hz, actual freq is 250.0004 Hz, precision is 2**10
ntp uptime is 590700 (1/100 of seconds), resolution is 4000
reference time is E6475AA9.46A7F060 (19:39:05.276 UTC Sun Jun 5 2022)
clock offset is -3.4704 msec, root delay is 1.10 msec
root dispersion is 45.22 msec, peer dispersion is 3.59 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000001804 s/s
system poll interval is 256, last update was 1402 sec ago.
```

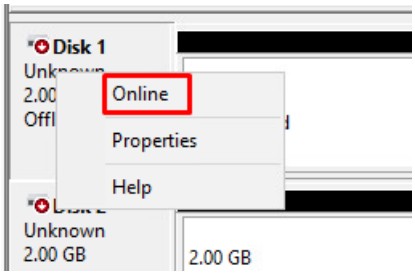
```
RTR-R#show ntp status
Clock is synchronized, stratum 6, reference is 192.168.100.200
nominal freq is 250.0000 Hz, actual freq is 249.9999 Hz, precision is 2**10
ntp uptime is 585600 (1/100 of seconds), resolution is 4016
reference time is E6475AB3.18937500 (19:39:15.096 UTC Sun Jun 5 2022)
clock offset is -3.6175 msec, root delay is 2.09 msec
root dispersion is 77.96 msec, peer dispersion is 4.57 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000237 s/s
system poll interval is 256, last update was 1426 sec ago.
```

17. Настройка RAID-1 на SRV

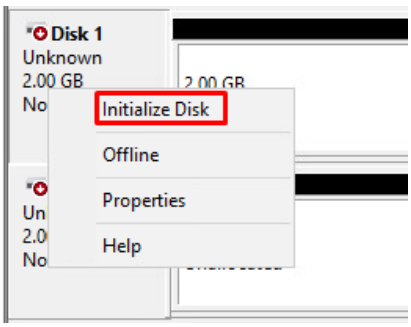
Переходим в настройки дисков:

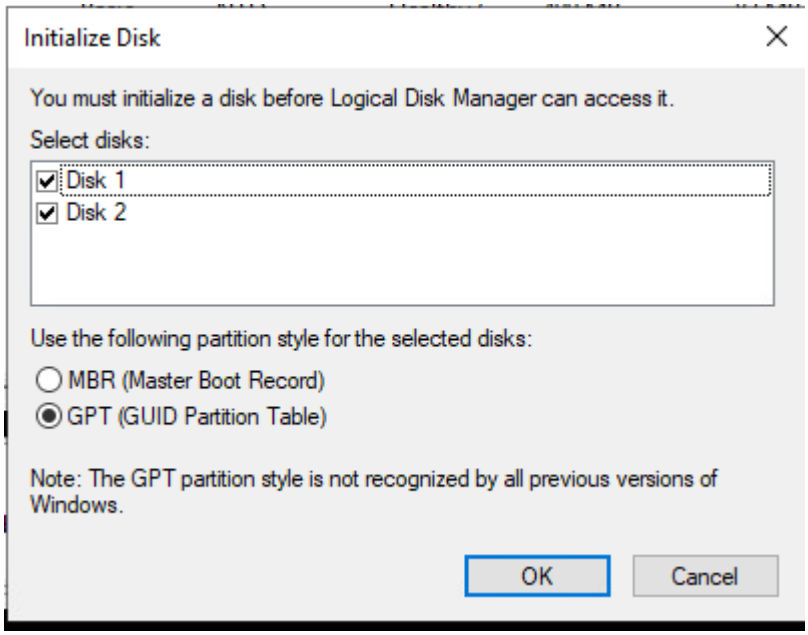


Переводим два диска в режим Online:

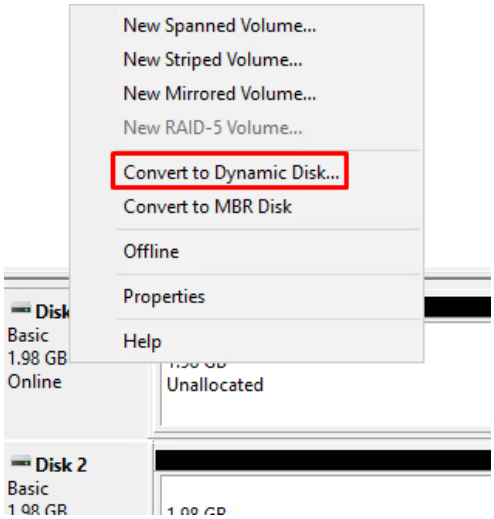


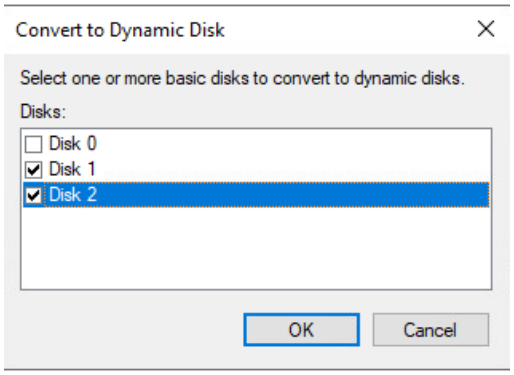
Инициализируем диски:



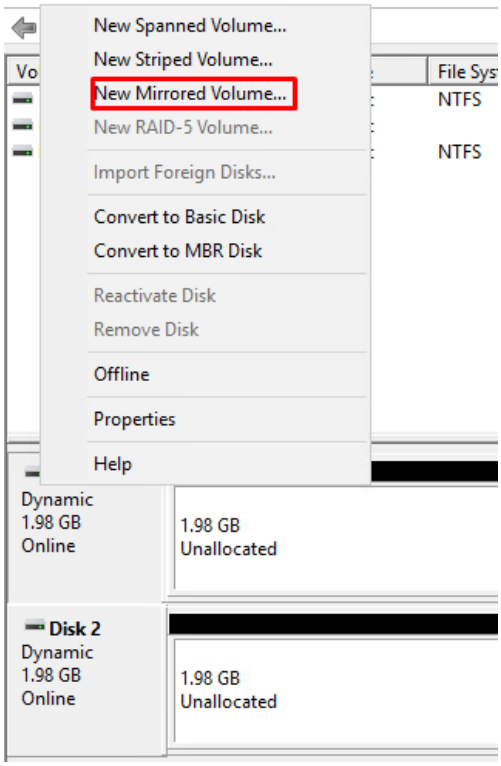


Конвертируем диски в динамические:

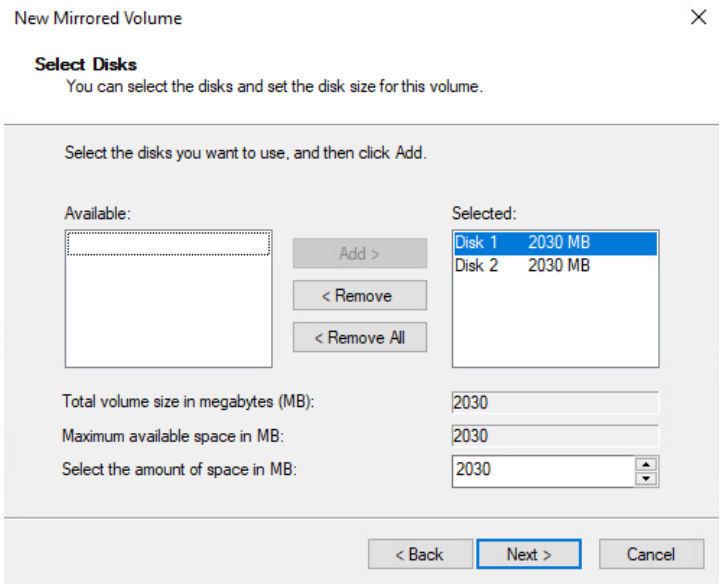




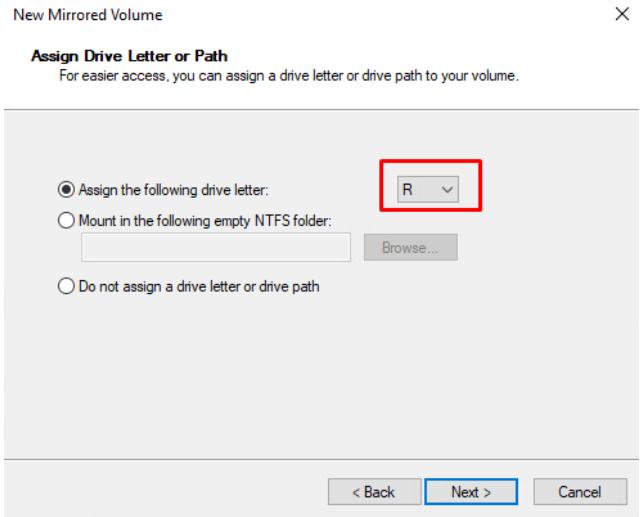
Создаем RAID-1 (Тип: Зеркало – это по заданию):



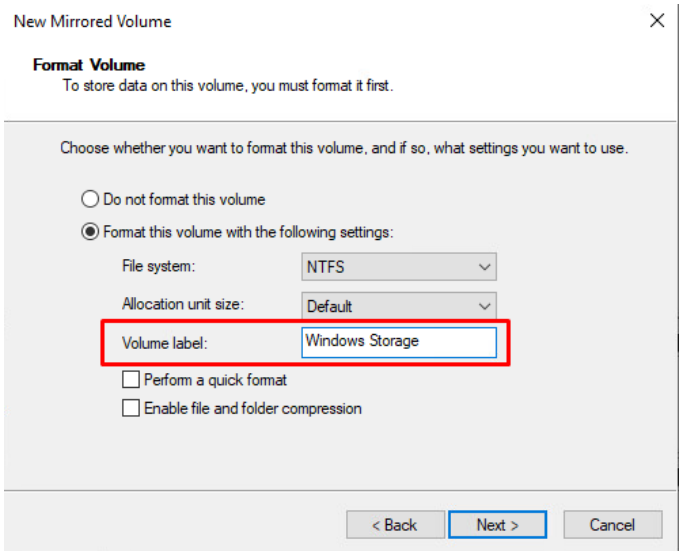
Выбираем оба диска, чтоб было как на скрине:



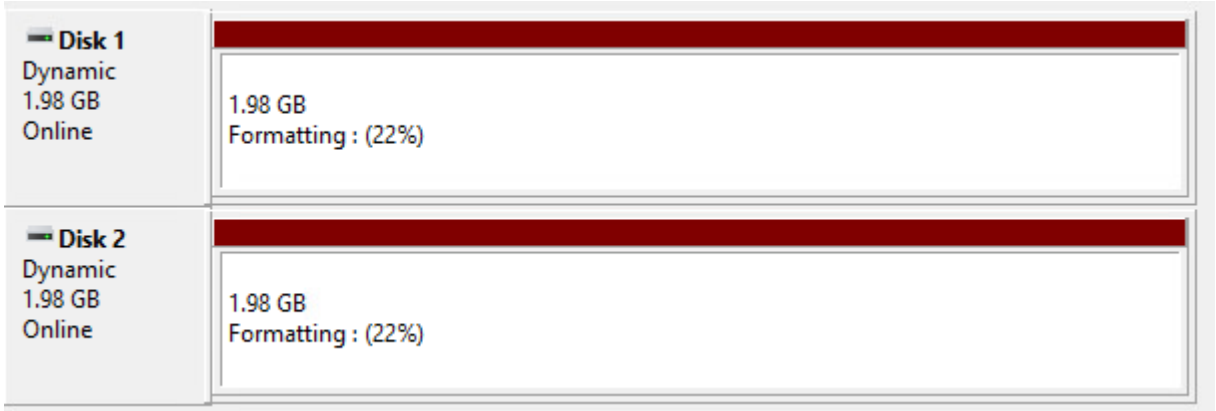
Назначаем букву согласно заданию (в нашем случае R:\):



Даем название, можно как у меня:

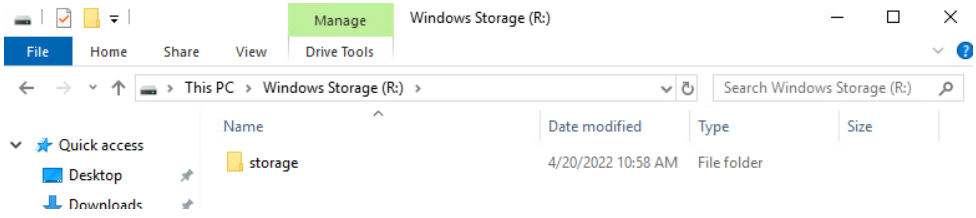


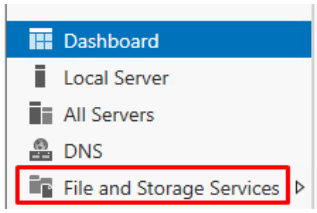
Ждем окончания операции:

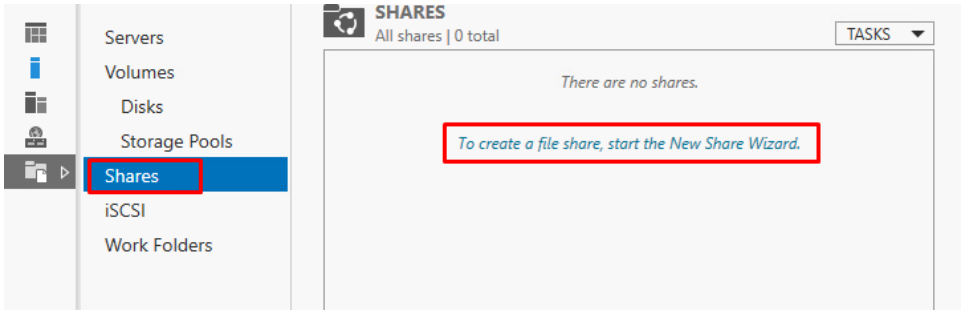


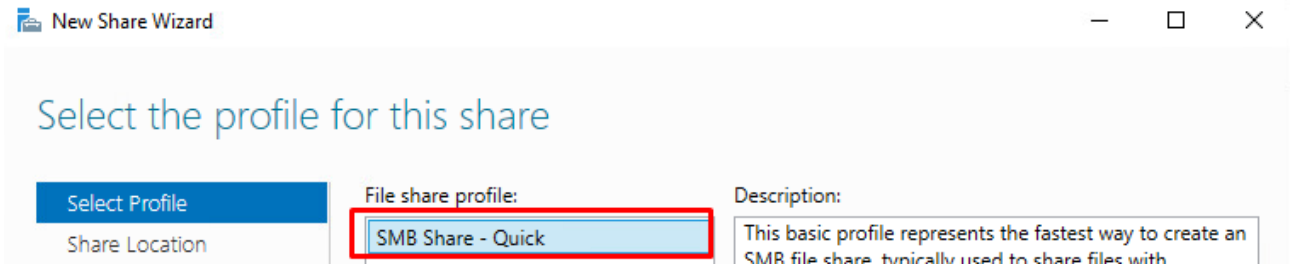
18. Настраиваем Samba-сервер на машине SRV:

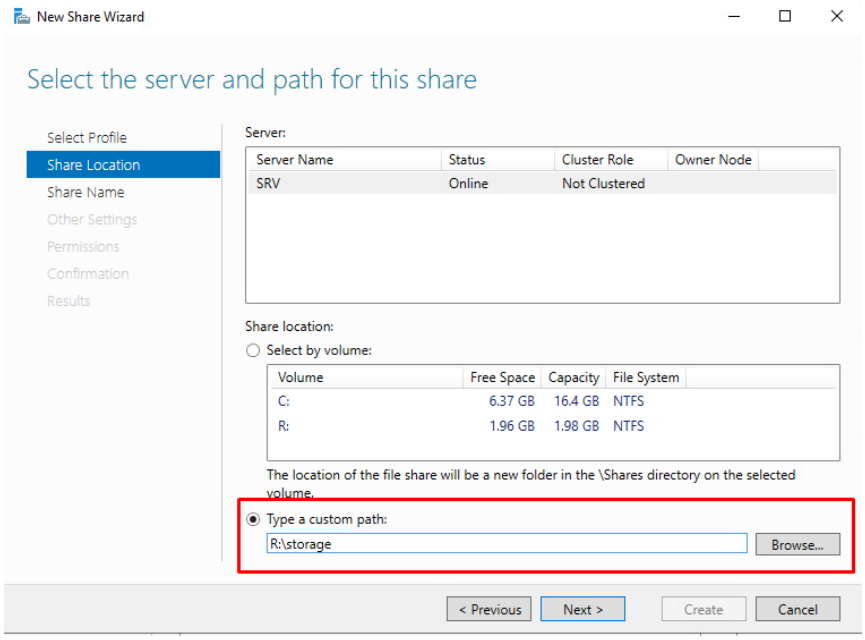
Настраиваем общую папку:

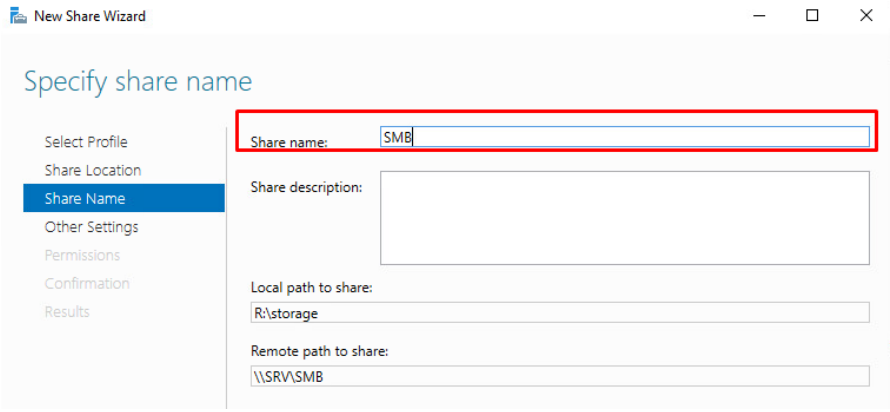

Manage
Windows Storage (R:)
File
Home
Share
View
Drive Tools
This PC › Windows Storage (R:) ›
Search Windows Storage (R:)
Name
Date modified
Type
Size
Quick access
Desktop
Downloads
storage
4/20/2022 10:58 AM
File folder

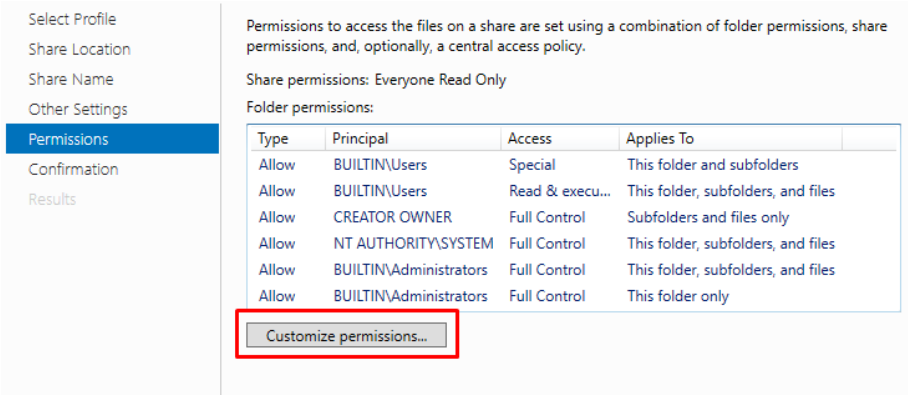

Dashboard
Local Server
All Servers
DNS
File and Storage Services

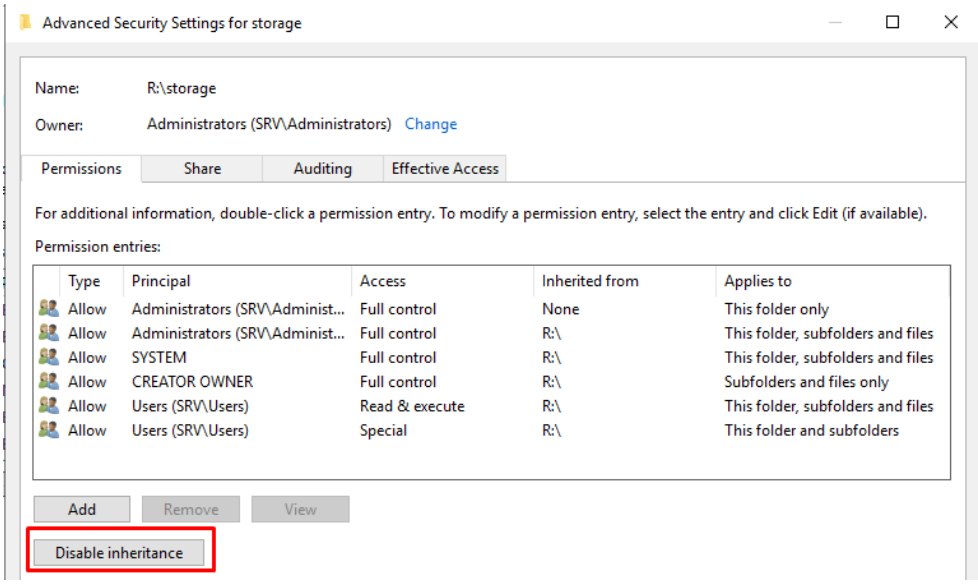

Servers
Volumes
Disks
Storage Pools
Shares
iSCSI
Work Folders
SHARES
All shares | 0 total
TASKS
There are no shares.
To create a file share, start the New Share Wizard.

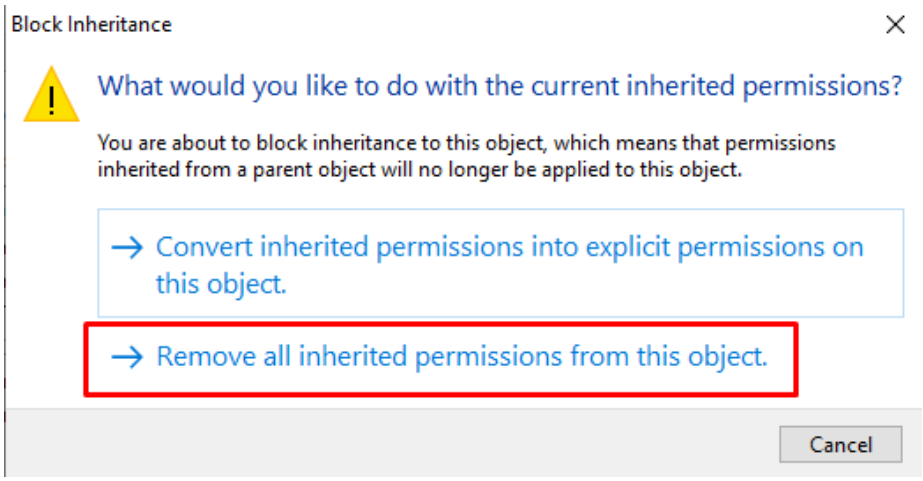

New Share Wizard
Select the profile for this share
Select Profile
Share Location
File share profile:
SMB Share - Quick
Description:
This basic profile represents the fastest way to create an


New Share Wizard
Select the server and path for this share
Select Profile
Share Location
Share Name
Other Settings
Permissions
Confirmation
Results
Server:
Server Name
Status
Cluster Role
Owner Node
SRV
Online
Not Clustered
Share location:
Select by volume:
Volume
Free Space
Capacity
File System
C:
6.37 GB
16.4 GB
NTFS
R:
1.96 GB
1.98 GB
NTFS
The location of the file share will be a new folder in the \Shares directory on the selected volume.
Type a custom path:
R:\storage
Browse...
< Previous
Next >
Create
Cancel


New Share Wizard
Specify share name
Select Profile
Share Location
Share Name
Other Settings
Permissions
Confirmation
Results
Share name:
SMB
Share description:
Local path to share:
R:\storage
Remote path to share:
\\SRV\SMB


Select Profile
Share Location
Share Name
Other Settings
Permissions
Confirmation
Results
Permissions to access the files on a share are set using a combination of folder permissions, share permissions, and, optionally, a central access policy.
Share permissions: Everyone Read Only
Folder permissions:
Type Principal Access Applies To
Allow BUILTIN\Users Special This folder and subfolders
Allow BUILTIN\Users Read & execu... This folder, subfolders, and files
Allow CREATOR OWNER Full Control Subfolders and files only
Allow NT AUTHORITY\SYSTEM Full Control This folder, subfolders, and files
Allow BUILTIN\Administrators Full Control This folder, subfolders, and files
Allow BUILTIN\Administrators Full Control This folder only
Customize permissions...


Advanced Security Settings for storage
Name: R:\storage
Owner: Administrators (SRV\Administrators) Change
Permissions
Share
Auditing
Effective Access
For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).
Permission entries:
Type Principal Access Inherited from Applies to
Allow Administrators (SRV\Administ... Full control None This folder only
Allow Administrators (SRV\Administ... Full control R:\ This folder, subfolders and files
Allow SYSTEM Full control R:\ This folder, subfolders and files
Allow CREATOR OWNER Full control R:\ Subfolders and files only
Allow Users (SRV\Users) Read & execute R:\ This folder, subfolders and files
Allow Users (SRV\Users) Special R:\ This folder and subfolders
Add
Remove
View
Disable inheritance


Block Inheritance
What would you like to do with the current inherited permissions?
You are about to block inheritance to this object, which means that permissions inherited from a parent object will no longer be applied to this object.
Convert inherited permissions into explicit permissions on this object.
Remove all inherited permissions from this object.
Cancel

Permission entries:

Type	Principal	Access	Inherited from	Applies to
Allow	Administrators (SRV\Administ...	Full control	None	This folder only

Add | Remove | View

Permission Entry for storage

Principal: Select a principal

Type: Allow

Applies to: This folder, subfolders and files

Select User or Group

Select this object type:

User, Group, or Built-in security principal | Object Types...

From this location:

SRV | Locations...

Enter the object name to select (examples):

Everyone | Check Names

Advanced... | OK | Cancel

Permission Entry for storage

Principal: Everyone Select a principal

Type: Allow

Applies to: This folder, subfolders and files

Basic permissions:

Show advanced permissions

- [x] Full control
- [x] Modify
- [x] Read & execute
- [x] List folder contents
- [x] Read
- [x] Write
- [ ] Special permissions

- [ ] Only apply these permissions to objects and/or containers within this container

Clear all

Permissions | Share | Auditing | Effective Access

For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).

Permission entries:

Type	Principal	Access	Inherited from	Applies to
Allow	Administrators (SRV\Administ...	Full control	None	This folder, subfolders and files
Allow	Everyone	Full control	None	This folder, subfolders and files

Add | Remove | Edit

Permissions | Share | Auditing | Effective Access

To modify share permissions, select the entry and click Edit.

Network location for this share: \\SRV\SMB

Permission entries:

Type	Principal	Access
Allow	Everyone	Read

Add | Remove | Edit

Type: Allow

Permissions:

- [x] Full Control
- [x] Change
- [x] Read
- [ ] Special permissions

Clear all

Permission entries:

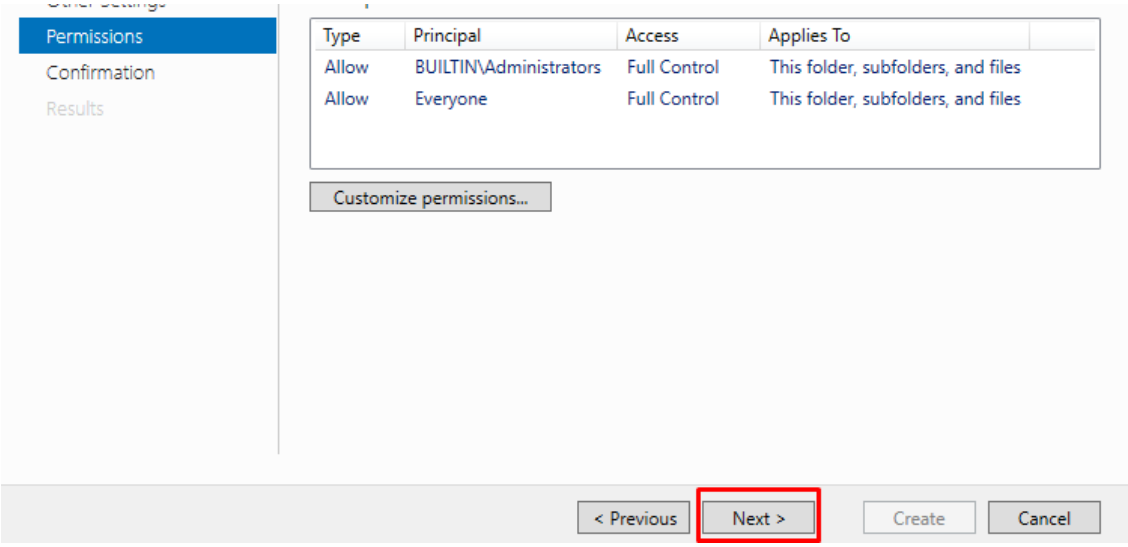
Type	Principal	Access	Inherited from	Applies to
Allow	Administrators (SRV\Administ...	Full control	None	This folder, subfolders and files
Allow	Everyone	Full control	None	This folder, subfolders and files

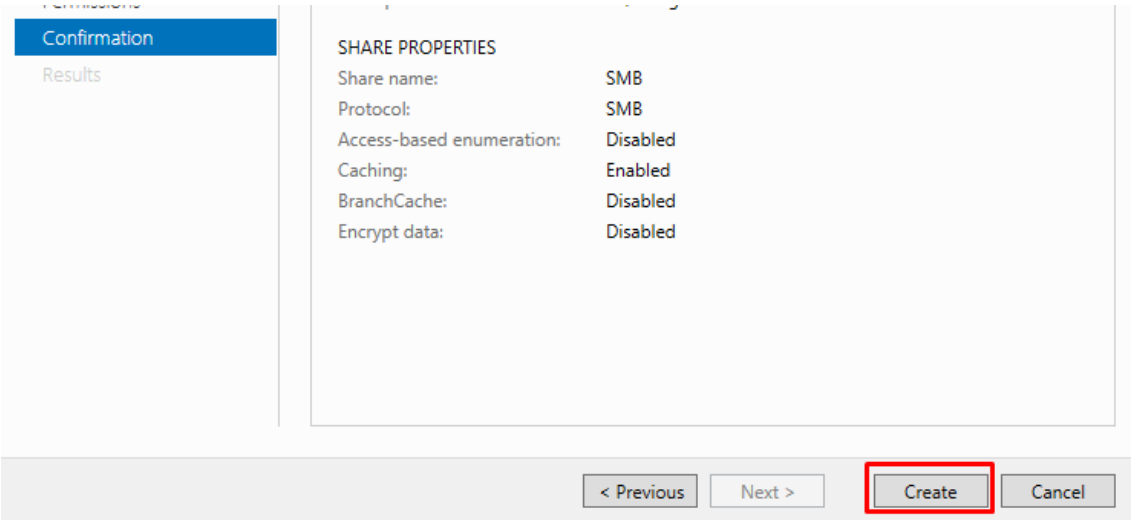
Add | Remove | Edit

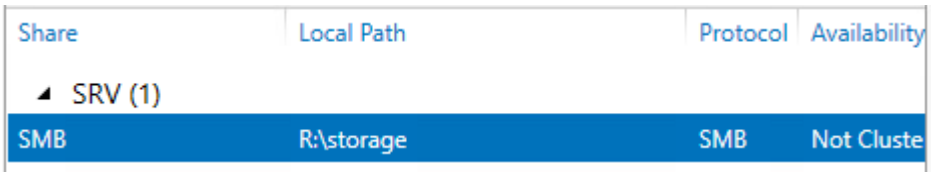
Enable inheritance

- [ ] Replace all child object permission entries with inheritable permission entries from this object

OK | Cancel | Apply







19.Настраиваем общую папку на машинах Web-L и Web-R

```
root@debian:~# apt install cifs-utils smbclient -y
```

Задаем имя пользователя и пароль для SMB-client:

```
root@debian:~# vim /root/.smbclient
```

```
username=Administrator
password=P@ssw0rd
```

Пароль ставится от учетной записи Администратора на машине SRV.

Добавляем папку в файл для smb:

```
root@debian:~# vim /etc/fstab
```

```
//srv.int.demo.wsr/smb  /opt/share cifs user,rw,_netdev,credentials=/root/.smbclient    0       0
```

Создаем каталог, который указали в файле:

```
root@debian:~# mkdir /opt/share
```

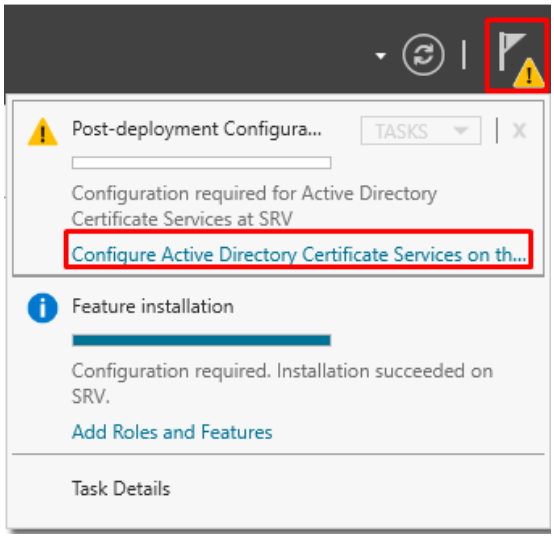
Монтируем общую папку:

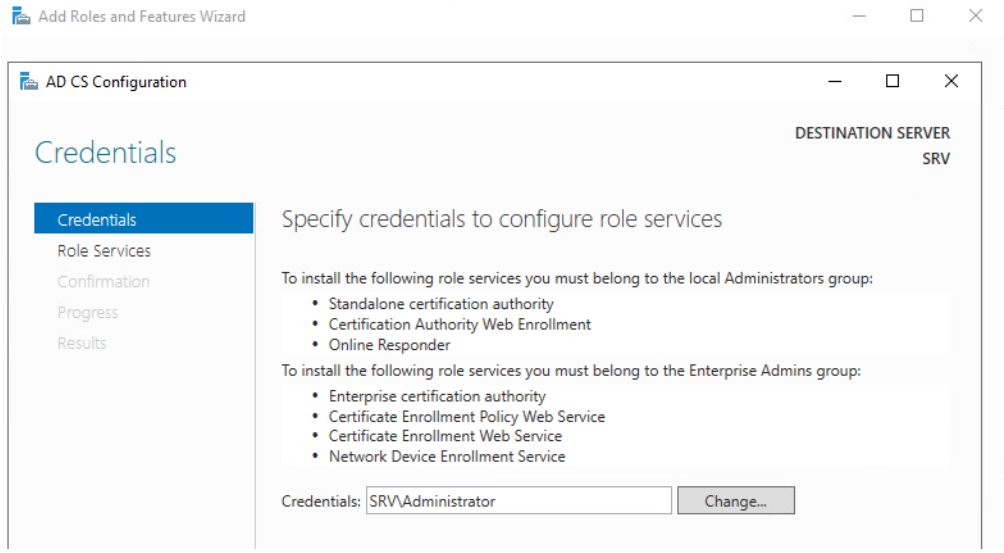
```
root@debian:~# mount -a
```

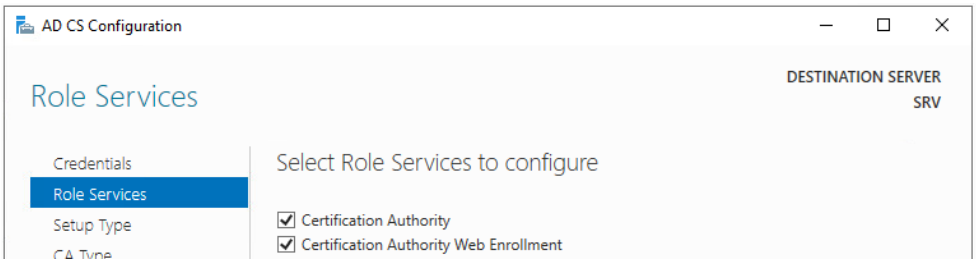
Делаем аналогично на Web-R.

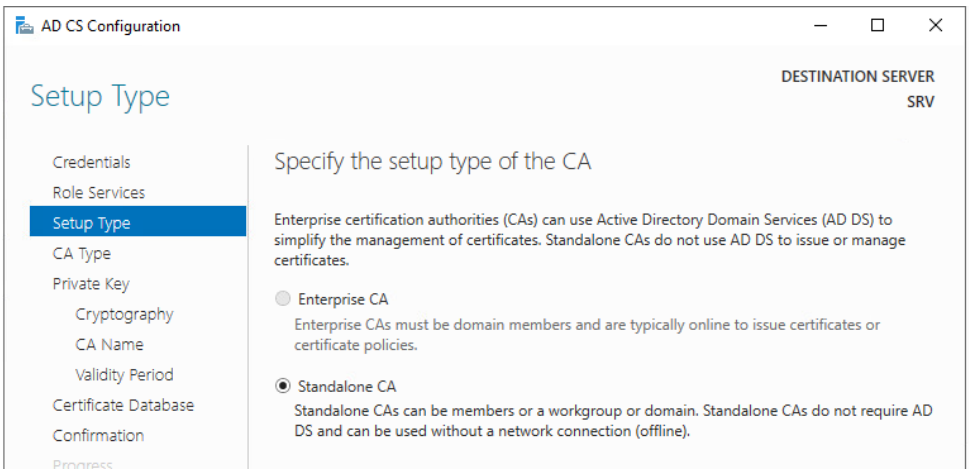
20. Настройка центра сертификации на машине SRV

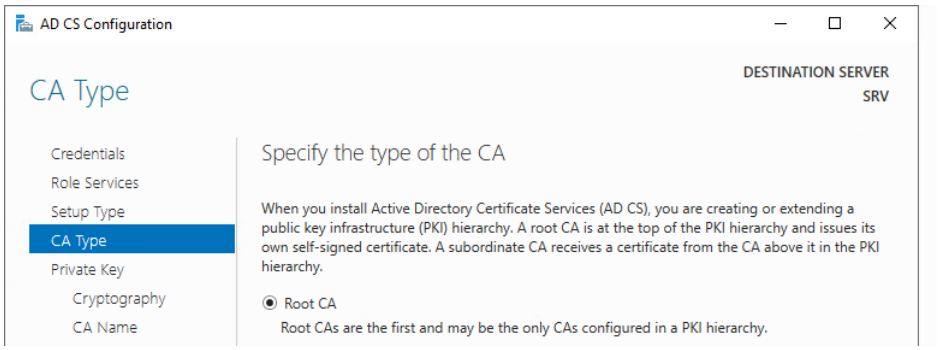
Создаем центр сертификации:

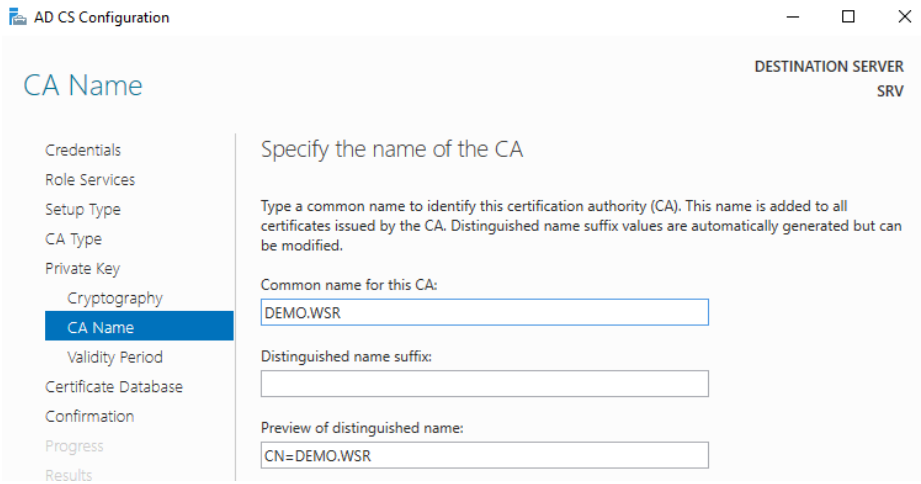








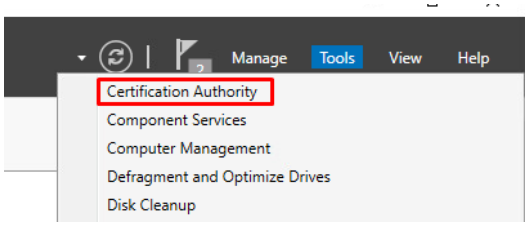


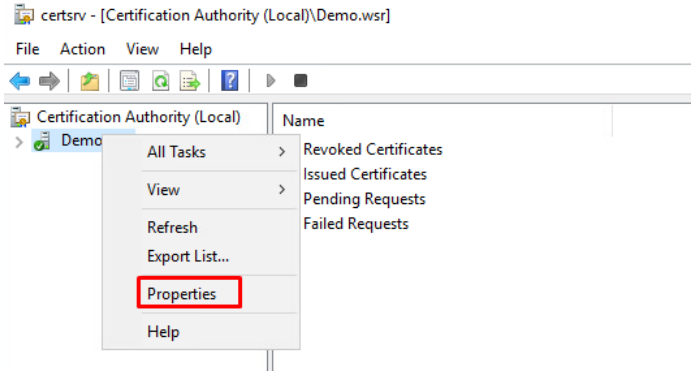


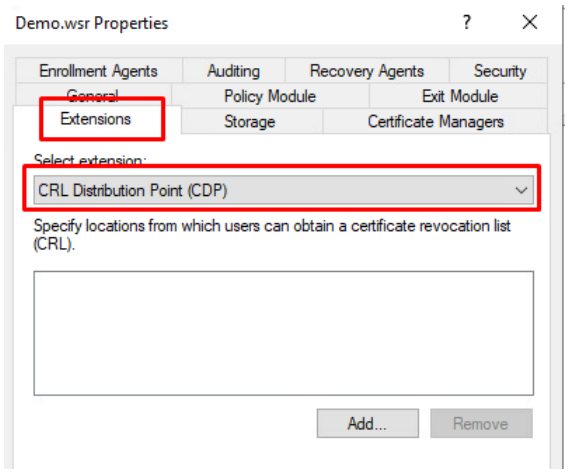
Заходим в PowerShellISE и прописываем там команды для создания и перемещения сертификатов:

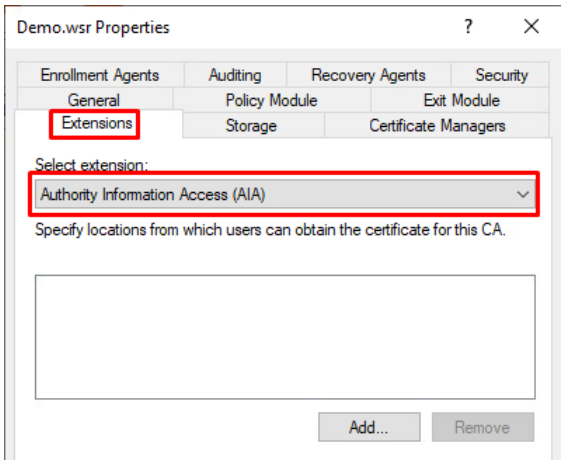
```
New-SelfSignedCertificate -Subject "localhost"
Get-ChildItem Cert:\LocalMachine\My
Move-Item Cert:\LocalMachine\My\CED99DC9B86A014B43AA624ABDDC5BE005F4109C -Destination Cert:\LocalMachine\WebHosting\
```

Заходим в Certification Authority и удаляем все их двух колонок, чтобы было все, как на скрине:



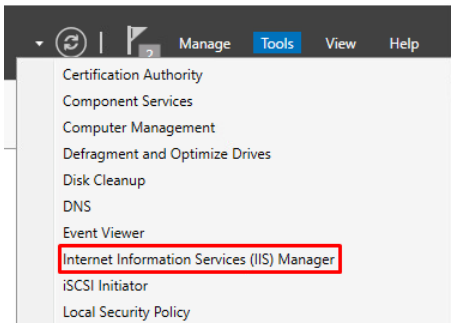


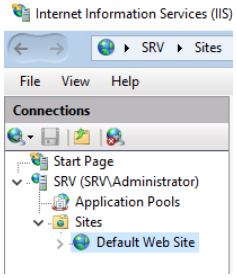


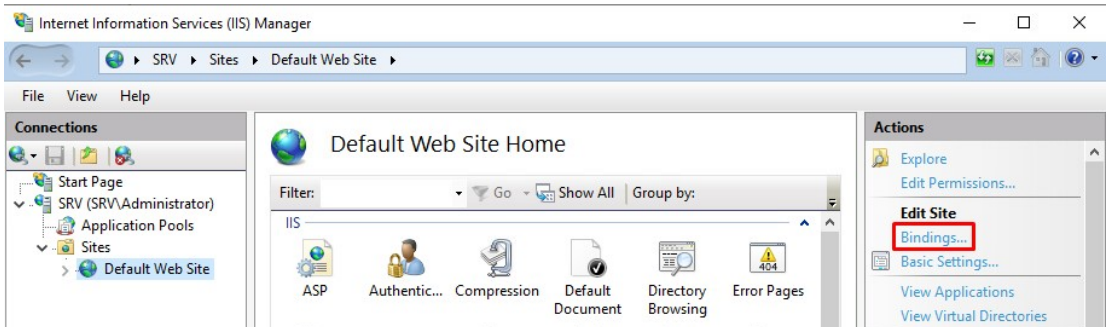


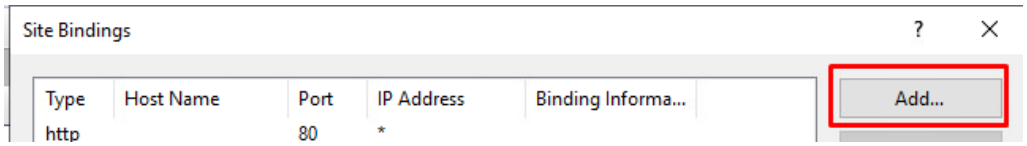
Далее жмем «ОК» и соглашаемся с перезагрузкой службы.

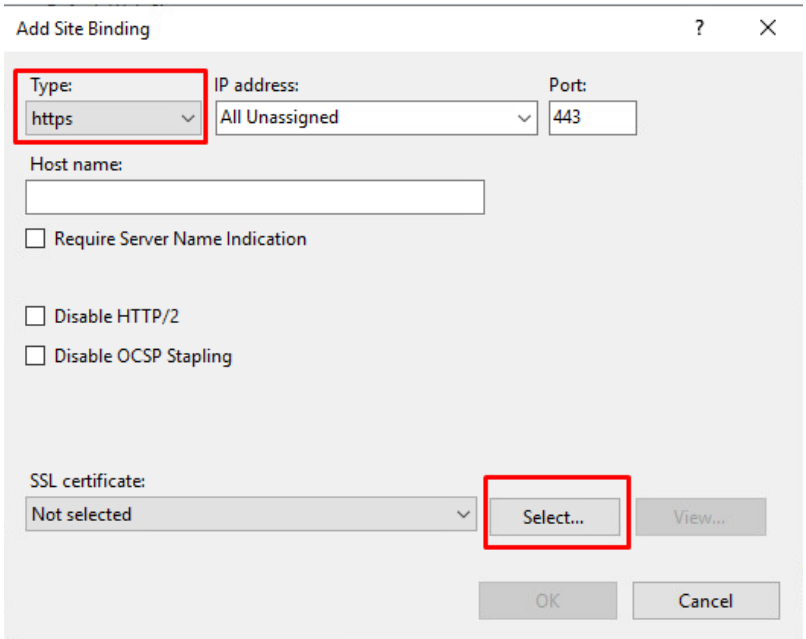
Настраиваем стандартный веб-сайт для работы по https по нашему сертификату:

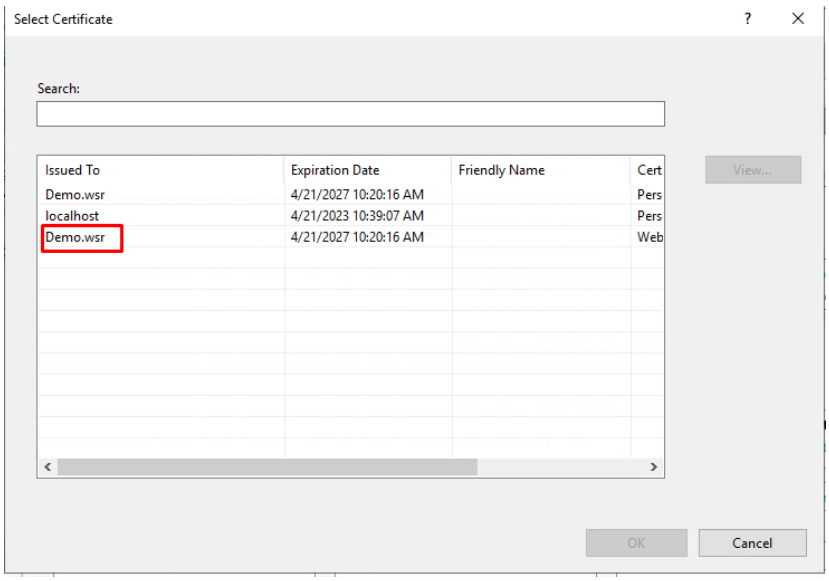


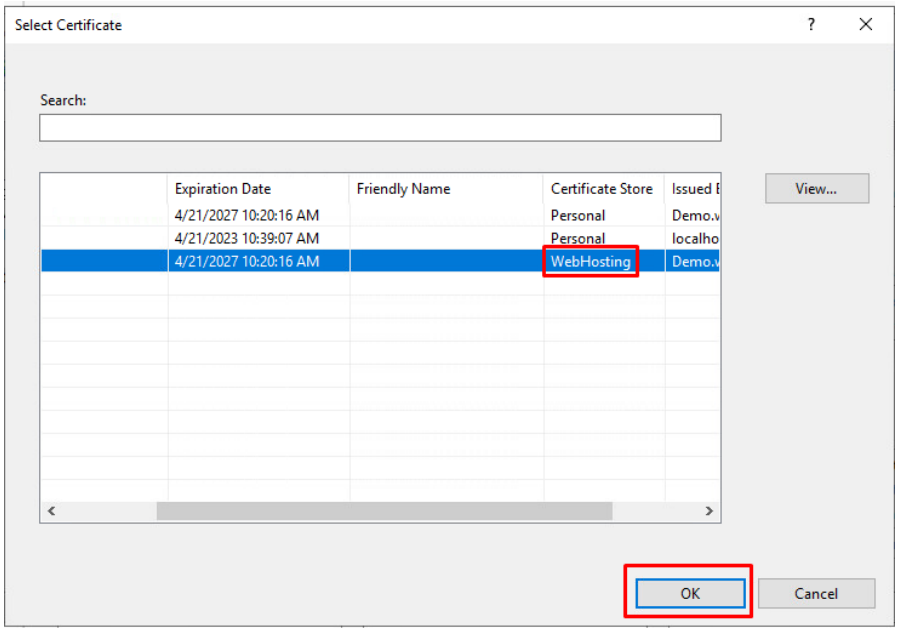

Internet Information Services (IIS)
SRV ▸ Sites
File View Help
Connections
Start Page
SRV (SRV\Administrator)
Application Pools
Sites
Default Web Site


Internet Information Services (IIS) Manager
SRV ▸ Sites ▸ Default Web Site ▸
File View Help
Connections
Start Page
SRV (SRV\Administrator)
Application Pools
Sites
Default Web Site
Default Web Site Home
Filter:
Go
Show All
Group by:
IIS
ASP
Authentic...
Compression
Default Document
Directory Browsing
Error Pages
Actions
Explore
Edit Permissions...
Edit Site
Bindings...
Basic Settings...
View Applications
View Virtual Directories

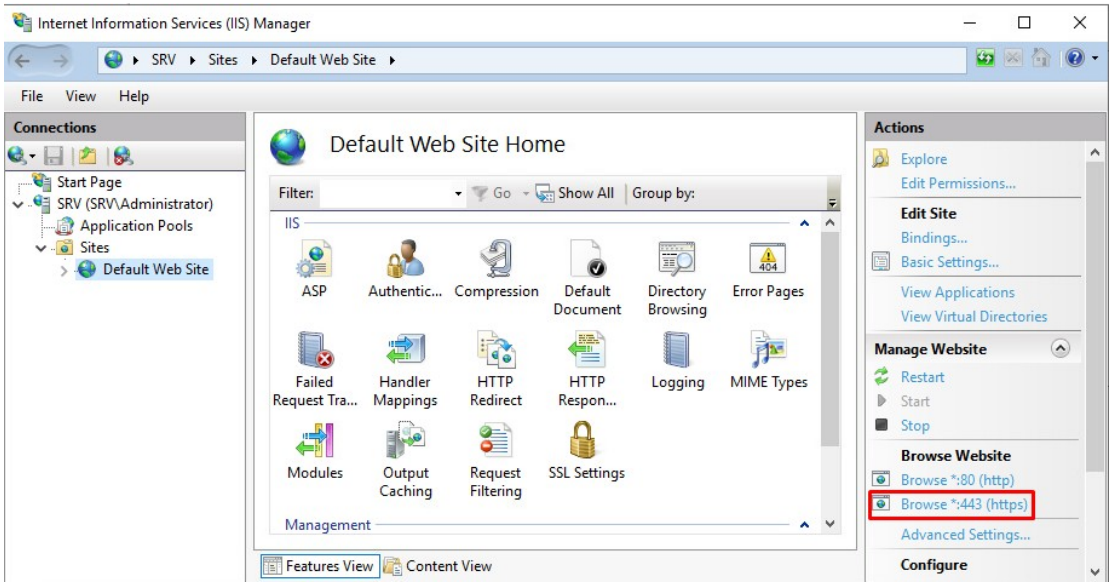

Site Bindings
Type
Host Name
Port
IP Address
Binding Informa...
http
80
*
Add...


Add Site Binding
Type:
https
IP address:
All Unassigned
Port:
443
Host name:
Require Server Name Indication
Disable HTTP/2
Disable OCSP Stapling
SSL certificate:
Not selected
Select...
View...
OK
Cancel



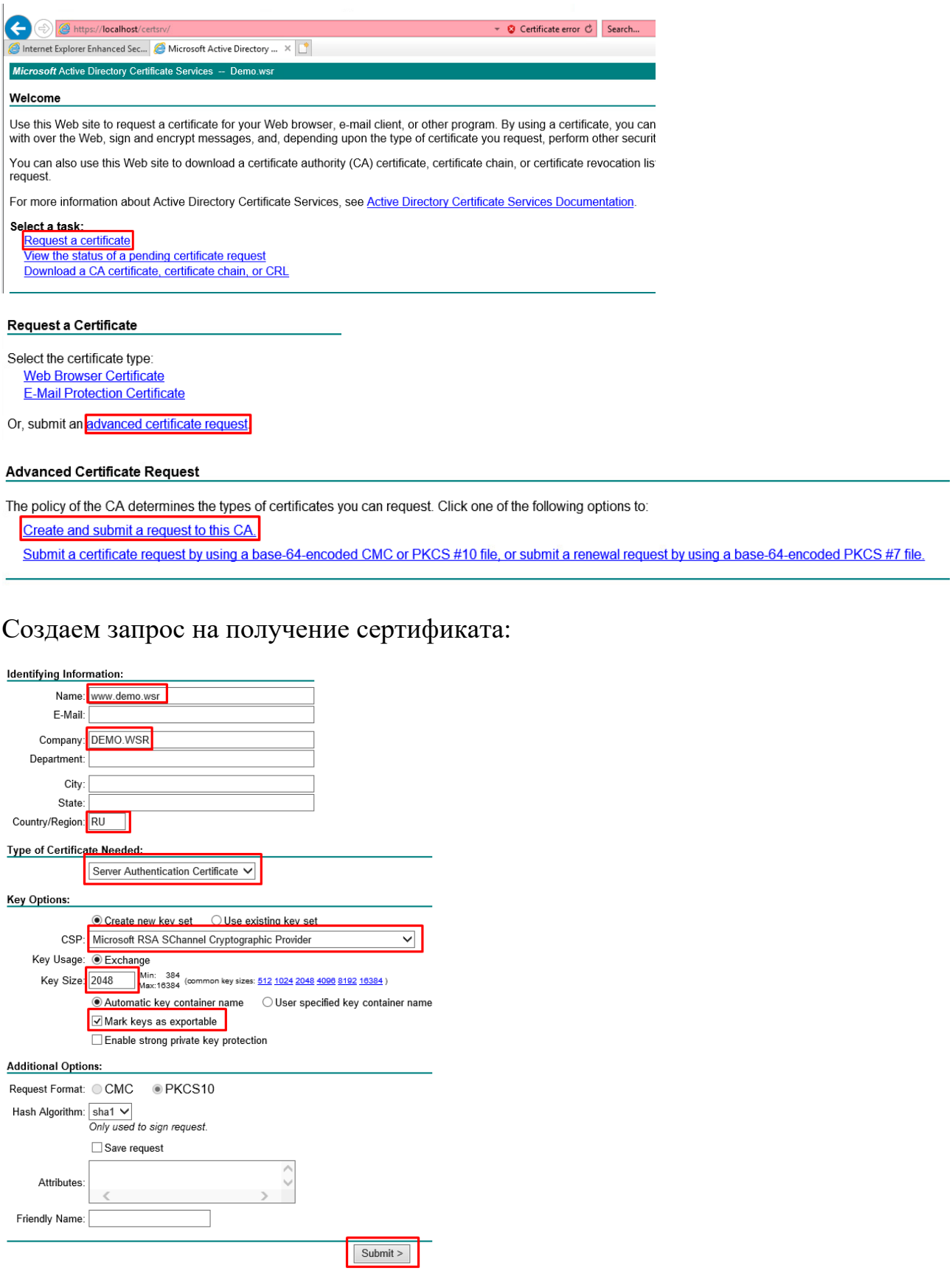


Должен появиться порт 443:

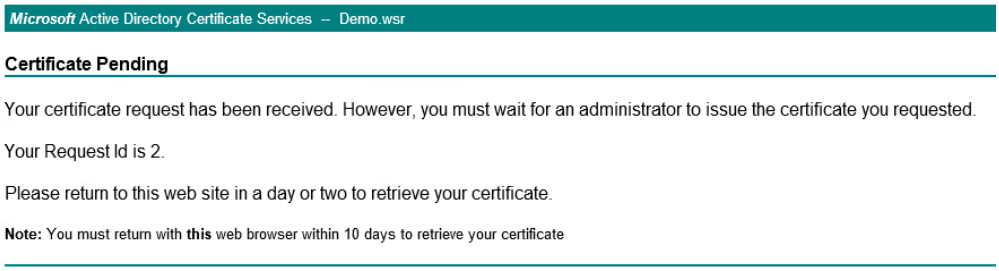


21.Настройка веб-приложения (Часть 1. Получение сертификата)

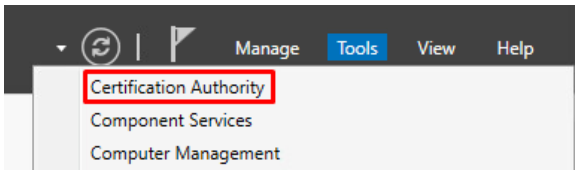
В браузере на машине SRV переходим по адресу https://localhost/certsrv для создания и получения сертификата:

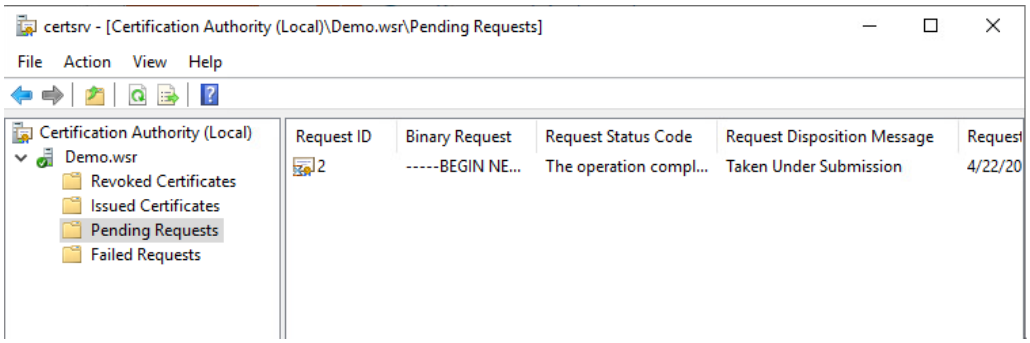


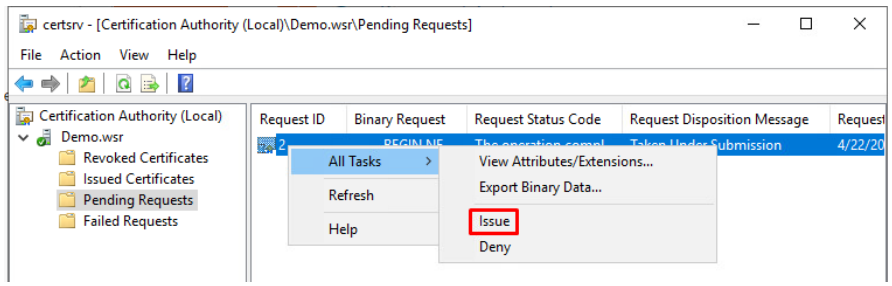
Создаем запрос на получение сертификата:



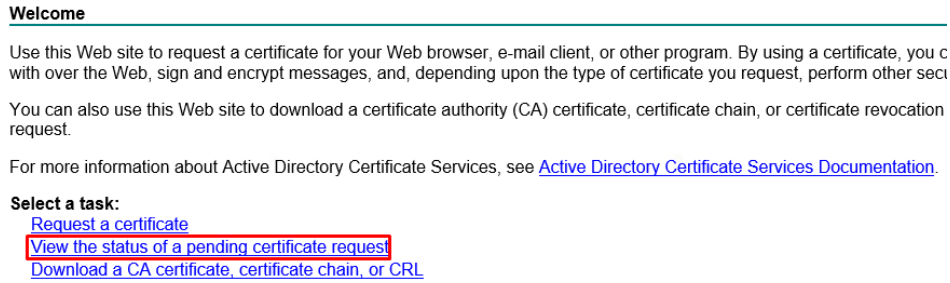
Одобряем сертификат в нашем центре сертификации:

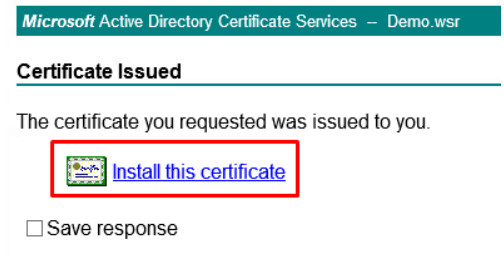


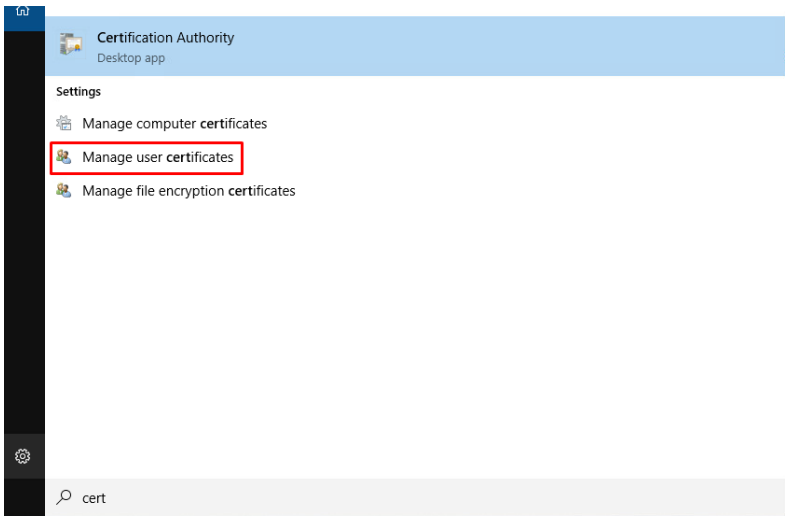




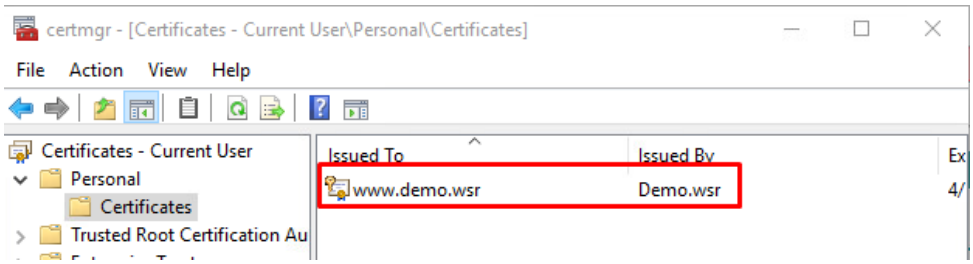
Устанавливаем сертификат:

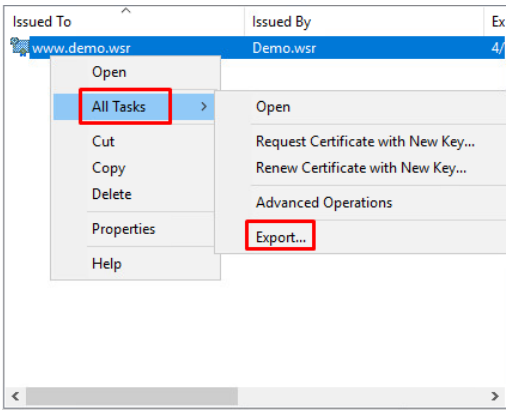


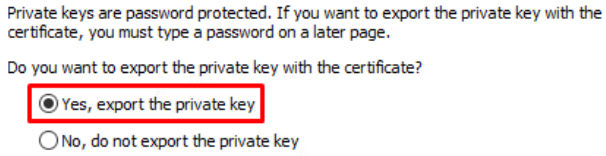


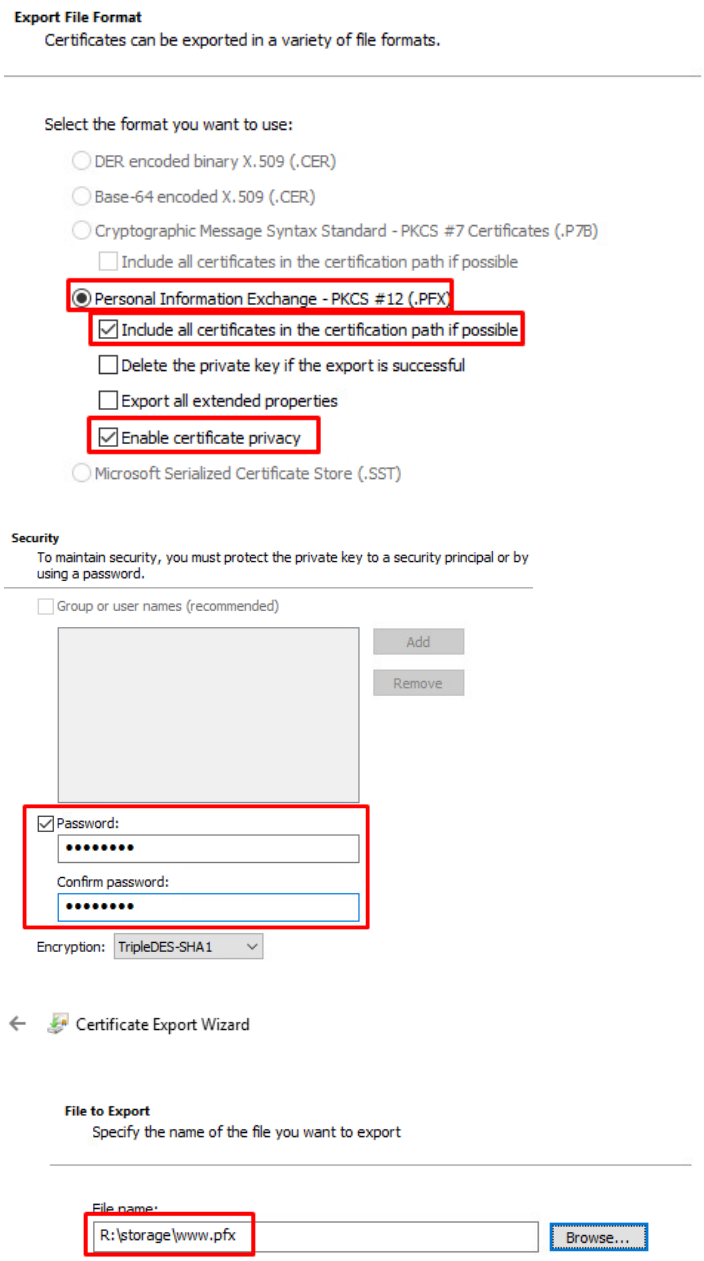


Экспортируем новый сертификат вместе с закрытым ключом:



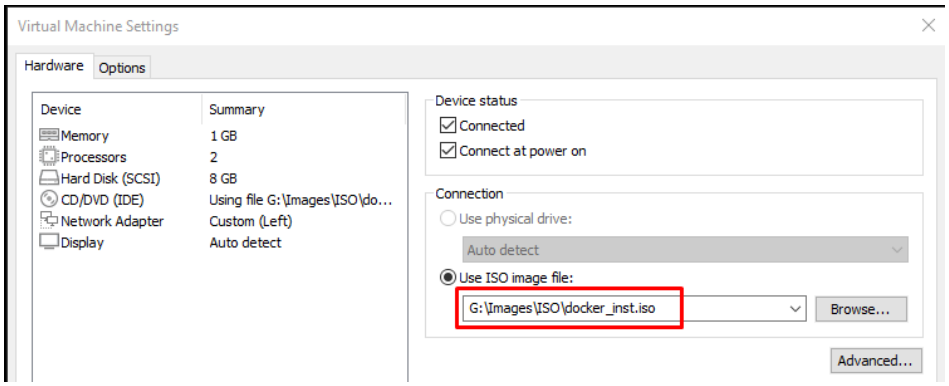






22. Устанавливаем приложение из контейнера Docker на машины WEB

Добавляем в дисковод диск с образом Docker:



Создаем каталог для монтирования диска с докером:

```
root@WEB-L:~# mkdir /opt/docker
```

```
root@WEB-R:~# mkdir /opt/docker
```

Монтируем диск с докером в созданный каталог:

```
root@WEB-L:~# mount /dev/sr0 /opt/docker/
mount: /opt/docker: WARNING: source write-protected, mounted read-only.
```

```
root@WEB-R:~# mount /dev/sr0 /opt/docker/
mount: /opt/docker: WARNING: source write-protected, mounted read-only.
```

Смотрим, что находится сейчас в каталоге:

```
root@WEB-L:~# ls /opt/docker/
appdocker0.zip
containerd.io_1.4.12-1_amd64.deb
dbus-user-session_1.12.20-2_amd64.deb
docker-ce_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-ce-cli_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-ce-rootless-extras_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-scan-plugin_0.12.0~debian-bullseye_amd64.deb
iptables_1.8.7-1_amd64.deb
Readme.txt
```

```
root@WEB-R:~# ls /opt/docker/
appdocker0.zip
containerd.io_1.4.12-1_amd64.deb
dbus-user-session_1.12.20-2_amd64.deb
docker-ce_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-ce-cli_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-ce-rootless-extras_5%253a20.10.12~3-0~debian-bullseye_amd64.deb
docker-scan-plugin_0.12.0~debian-bullseye_amd64.deb
iptables_1.8.7-1_amd64.deb
Readme.txt
```

Создаем каталог, чтобы скопировать файлы установки докера. Это нужно, чтобы мы смогли его установить, т.к. для установки также нужен диск Debian BD1, а в дисководе только диск с докером:

```
root@WEB-L:~# mkdir /files/
```

```
root@WEB-R:~# mkdir /files/
```

Создаем каталог, чтобы скопировать файл самого приложения для докера и файла Readme.txt:

```
root@WEB-L:~# mkdir /webapp/
```

```
root@WEB-R:~# mkdir /webapp/
```

Копируем установочные файлы докера в созданный нами каталог /files/:

```
root@WEB-L:~# cp /opt/docker/ip* /opt/docker/d* /opt/docker/con* /files/
```

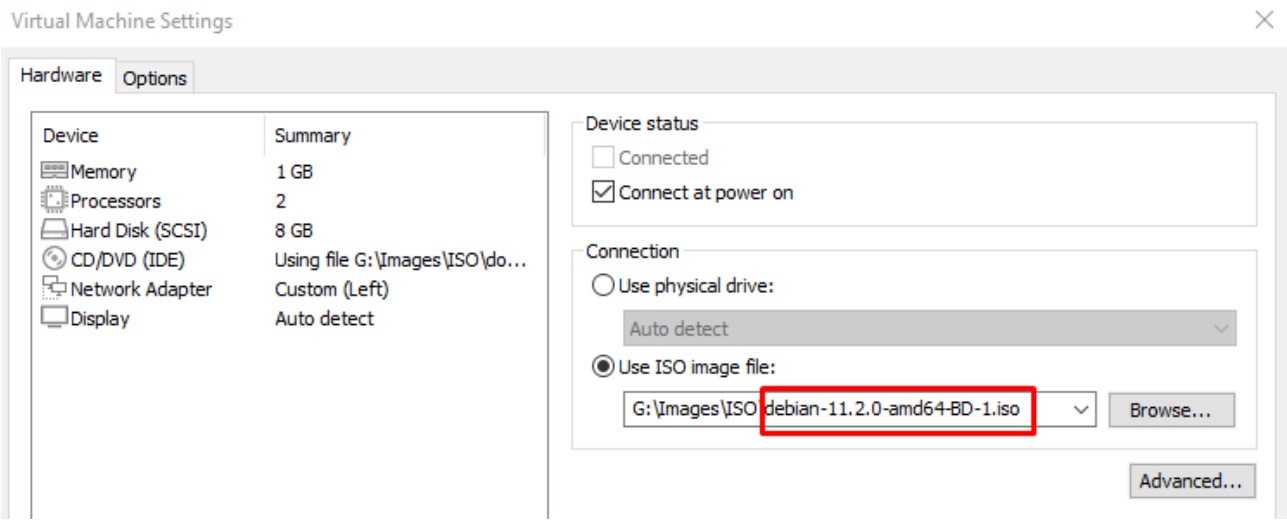
```
root@WEB-R:~# cp /opt/docker/ip* /opt/docker/d* /opt/docker/con* /files/
```

Копируем файл приложения и файл Readme.txt в созданный нами каталог /webapp/:

```
root@WEB-L:~# cp /opt/docker/appdocker0.zip /opt/docker/Readme.txt /webapp/
```

```
root@WEB-R:~# cp /opt/docker/appdocker0.zip /opt/docker/Readme.txt /webapp/
```

Выключаем машины и снова вставляем первый диск в дисковод:



Включаем машины и устанавливаем докер:

```
root@WEB-L:~# apt install /files/* -y
```

```
root@WEB-R:~# apt install /files/* -y
```

Загружаем приложение в докер:

```
root@WEB-L:~# docker load < /webapp/appdocker0.zip
e81bff2725db: Loading layer  72.52MB/72.52MB
9d1100648d6d: Loading layer  7.377MB/7.377MB
da27cd6210af: Loading layer  29.37MB/29.37MB
c511542be34c: Loading layer  4.608kB/4.608kB
e8acaef3fb23: Loading layer  10.07MB/10.07MB
c1feea9acfbd: Loading layer    200MB/200MB
3faaba577d8d: Loading layer  22.53kB/22.53kB
1149d2cd8b5d: Loading layer  42.48MB/42.48MB
13705d5c900f: Loading layer   2.56kB/2.56kB
Loaded image: appdocker0:latest
```

```
root@WEB-R:~# docker load < /webapp/appdocker0.zip
e81bff2725db: Loading layer  72.52MB/72.52MB
9d1100648d6d: Loading layer  7.377MB/7.377MB
da27cd6210af: Loading layer  29.37MB/29.37MB
c511542be34c: Loading layer  4.608kB/4.608kB
e8acaef3fb23: Loading layer  10.07MB/10.07MB
c1feea9acfbd: Loading layer    200MB/200MB
3faaba577d8d: Loading layer  22.53kB/22.53kB
1149d2cd8b5d: Loading layer  42.48MB/42.48MB
13705d5c900f: Loading layer   2.56kB/2.56kB
Loaded image: appdocker0:latest
```

Проверяем, загрузилось ли приложение:

```
root@WEB-L:~# docker images
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
appdocker0   latest    1b842672420e   4 months ago   352MB
```

```
root@WEB-R:~# docker images
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
appdocker0   latest    1b842672420e   4 months ago   352MB
```

Запускаем приложение с именем webapp и пробросом портов с 8080 на 5000, потому что в файле Readme.txt сказано, что приложение работает на порте 5000:

```
root@WEB-L:~# docker run --name webapp -p 8080:5000 -d appdocker0
```

```
root@WEB-R:~# docker run --name webapp -p 8080:5000 -d appdocker0
```

Проверяем, успешно ли запустилось приложение:

```
root@WEB-L:~# docker ps
CONTAINER ID   IMAGE        COMMAND        CREATED          STATUS         PORTS
                   NAMES
112beb904a1f   appdocker0   "./start.sh"   10 seconds ago   Up 8 seconds   0.0.0.0:8080->5000/tcp, :
::8080->5000/tcp   webapp
```

```
root@WEB-R:~# docker ps
CONTAINER ID   IMAGE        COMMAND        CREATED          STATUS          PORTS
                   NAMES
2c90f896b36a   appdocker0   "./start.sh"   27 seconds ago   Up 25 seconds   0.0.0.0:8080->5000/tcp,
:::8080->5000/tcp   webapp
```

Выключаем http-сервер на роутерах и пробрасываем порты:

```
RTR-L(config)#no ip http server
```

```
RTR-L(config)#no ip http secure-server
```

```
RTR-L(config)#ip nat inside source static tcp 192.168.100.100 80 4.4.4.100 80
RTR-L(config)#ip nat inside source static tcp 192.168.100.100 443_4.4.4.100 443
```

```
RTR-R(config)#no ip http server
```

```
RTR-R(config)#no ip http secure-server
```

```
RTR-R(config)#ip nat inside source static tcp 172.16.100.100 80 5.5.5.100 80
RTR-R(config)#ip nat inside source static tcp 172.16.100.100 443 5.5.5.100 443
```

23. Настройка отказоустойчивости приложений на машинах WEB

Устанавливаем веб-сервер nginx на обе машины:

```
root@WEB-L:~# apt install nginx -y
```

Настраиваем WEB-L:

```
root@WEB-L:~# cd /opt/share/
```

Извлекаем ключ из сертификата:

```
root@WEB-L:/opt/share# openssl pkcs12 -nodes -nocerts -in www.pfx -out www.key
Enter Import Password:
```

Извлекаем сертификат:

```
root@WEB-L:/opt/share# openssl pkcs12 -nodes -nokeys -in www.pfx -out www.cer
Enter Import Password:
```

Копируем сертификат и ключ в /etc/nginx:

```
root@WEB-L:/opt/share# cp www.cer /etc/nginx/www.cer
root@WEB-L:/opt/share# cp www.key /etc/nginx/www.key
```

Настраиваем файл для работы с нашим сертификатом и ключом:

```
root@WEB-L:/opt/share# vim /etc/nginx/snippets/snakeoil.conf
```

```
ssl_certificate /etc/nginx/www.crt;
ssl_certificate_key /etc/nginx/www.key;
ssl_protocols TLSv1 TLSv1.1 TLSv1.2;
```

Настраиваем обратное проксирование для нашего Docker-контейнера а также перенаправление с http на https:

```
root@WEB-L:/opt/share# vim /etc/nginx/sites-available/default
```

```
upstream backend {
server 192.168.100.100:8080 fail_timeout=25
server 172.16.100.100:8080 fail_timeout=25;
}

server {
        listen 443 ssl default_server;
        include snippets/snakeoil.conf;
        server_name www.demo.wsr;
        location / {
                proxy_pass http://backend;
        }
}

server {
        listen 80 default_server;
        server_name _;
        return 301 https://www.demo.wsr;
}
```

Перезагружаем веб-сервер:

```
root@WEB-L:/opt/share# systemctl reload nginx
```

Копируем данные файлы в общую папку, чтобы передать их на машину WEB-R:

```
root@WEB-L:~# cp /etc/nginx/snippets/snakeoil.conf /opt/share/
root@WEB-L:~# cp /etc/nginx/sites-available/default /opt/share/
```

Закидываем данные файлы на машину WEB-R, не забудьте скопировать сертификаты:

```
root@WEB-R:~# cd /opt/share/
```

```
root@WEB-R:/opt/share# cp www.key www.crt /etc/nginx/
```

```
root@WEB-R:/opt/share# cp snakeoil.conf /etc/nginx/snippets/
```

```
root@WEB-R:/opt/share# cp default /etc/nginx/sites-available/
```

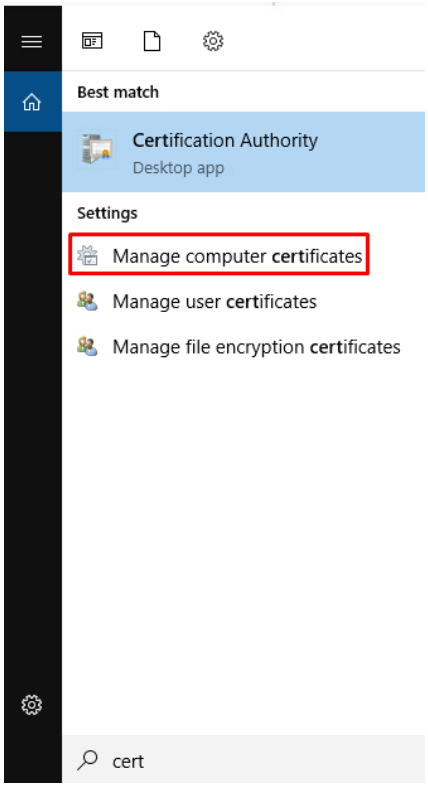
Перезагружаем веб-сервер:

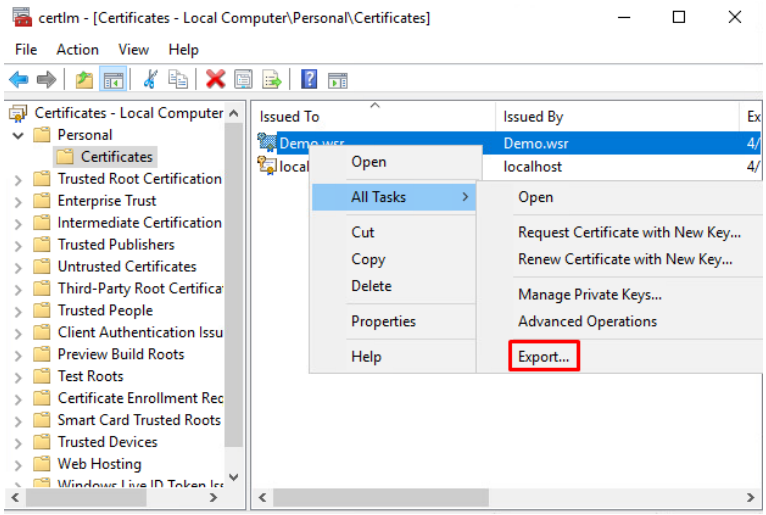
```
root@WEB-R:/opt/share# systemctl restart nginx
```

24. Добавляем основной сертификат на машину CLI и проверяем работу

На машине SRV:

Экспортируем корневой сертификат:



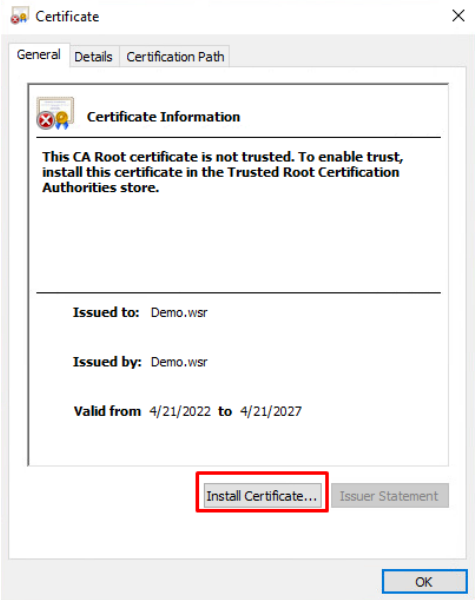


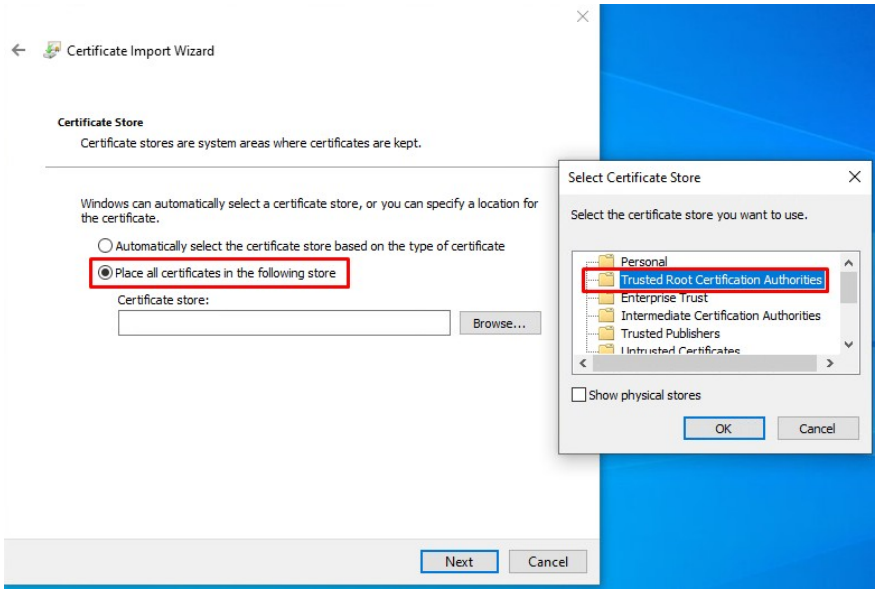
На машине CLI:

Передаем с помощью протокола scp корневой сертификат на машину CLI:

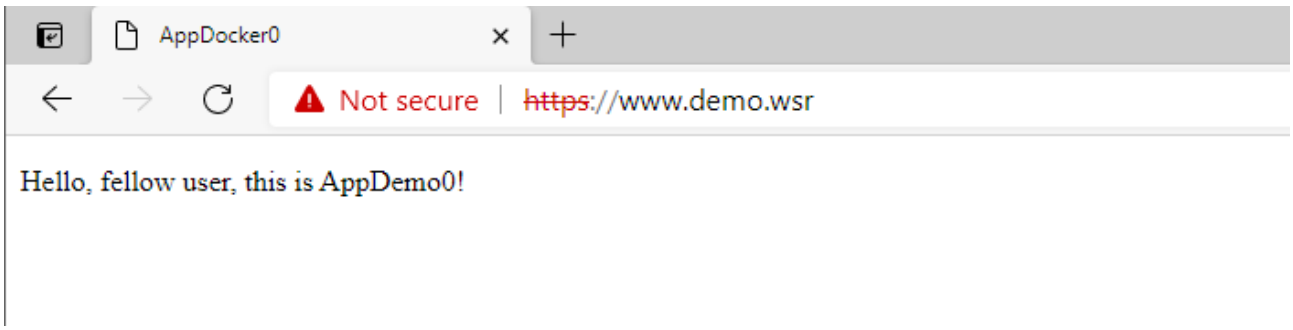
```
C:\Users\User>scp -P 2244 root@5.5.5.100:/opt/share/ca.cer C:\Users\User\Desktop
```

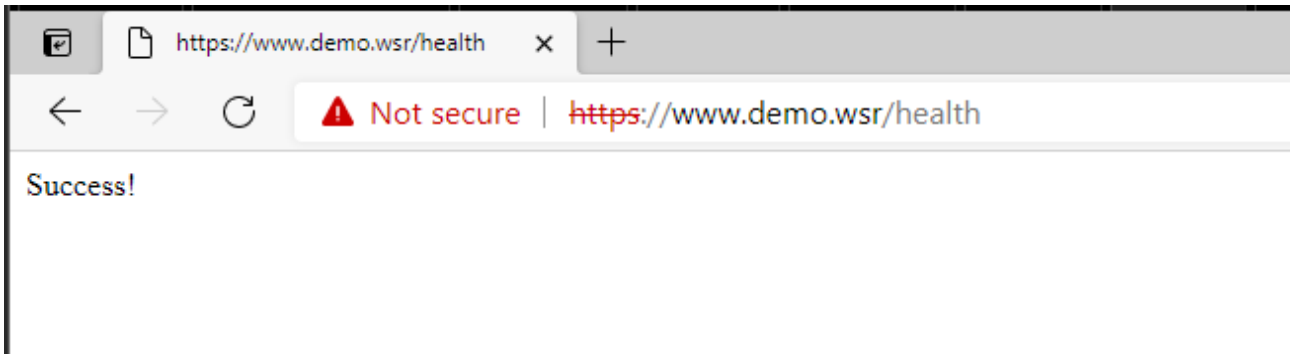
Устанавливаем сертификат в доверенное хранилище сертификатов:





Проверяем работоспособность на машине CLI:





Более подробная инструкция по работе с приложением находится в файле Readme.txt, находящемся на диске с докером.

На этом выполнение демоэкзамена закончено!!!